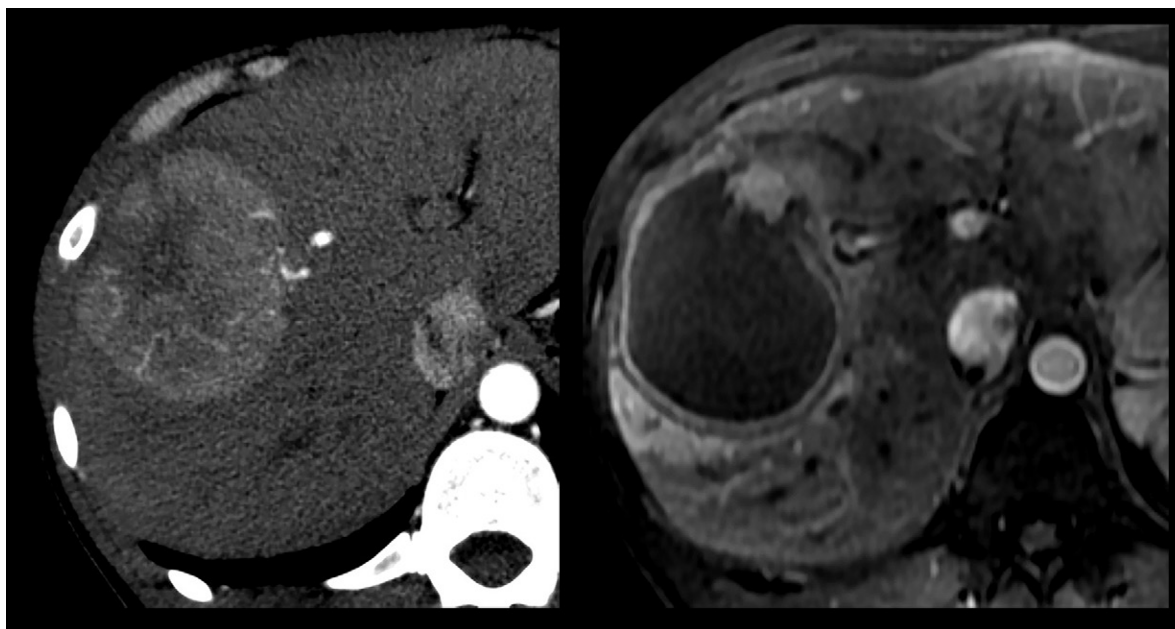


CT and MRI LI-RADS Treatment Response Assessment 2024: Core Concepts for Clinical Practice

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For management of hepatocellular carcinoma (HCC), locoregional therapies are commonly used as curative interventions, bridging strategies before a liver transplant, or methods to reduce the tumor burden. After treatment, imaging surveillance is essential to guiding subsequent patient treatment decisions by enabling timely detection of recurrence or residual tumor viability. However, imaging interpretation can be challenging due to variability in expected findings, based on the treatment modality, response achieved, and time since therapy. The CT and MRI Liver Imaging Reporting and Data System (LI-RADS) treatment response assessment algorithm was developed to standardize imaging interpretation and reporting after locoregional therapy for primary liver malignancies. Integrating emerging evidence and feedback from clinical users, the updated 2024 treatment response assessment algorithm establishes distinct criteria for nonradiation- and radiation-based therapies and includes ancillary MRI features to improve detection of tumor viability. The authors review the updated LI-RADS treatment response assessment algorithm, with particular attention to clinical applicability and implications for patient care.

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Introduction

Hepatocellular carcinoma (HCC) is the most common primary liver cancer and the sixth most frequently diagnosed cancer worldwide. Historically common in regions with a high prevalence of viral hepatitis, HCC is now diagnosed with increasing frequency in countries with a traditionally low risk of hepatitis, such as the United States. This increase reflects the effectiveness of antiviral treatments in reducing the number of cases of viral hepatitis along with an increase in alcohol-related cirrhosis and metabolic dysfunction-associated steatotic liver disease (1).

Established therapeutic strategies for patients with HCC include surgical resection, a liver transplant, locoregional therapy, and systemic therapy (2,3). Locoregional therapies are employed as curative interventions for early-stage disease, to reduce the tumor burden, or as a bridge therapy for patients waiting for a liver transplant. After locoregional therapy is performed, patients typically undergo imaging surveillance every 3 months with multiphasic CT or MRI. Accurate posttreatment imaging is essential for guiding patient treatment, enabling assessment of residual tumor viability, while allowing detection and staging of untreated lesions elsewhere in the liver.

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Abbreviations: ACR = American College of Radiology, HCC = hepatocellular carcinoma, LI-RADS = Liver Imaging Reporting and Data System, LR-TR = LI-RADS treatment response, SBRT = stereotactic body radiation therapy, TACE = transarterial chemoembolization, TR = treatment response, 3D = three-dimensional

TEACHING POINTS

- Incorporating emerging evidence and user feedback, the ACR updated the LI-RADS TR assessment criteria in 2024. This update introduced a dedicated algorithm for radiation-treated lesions, simplified the definition of viable disease, and incorporated ancillary imaging features to upgrade lesions to the viable category.
- LR-TR viable criteria were simplified in the 2024 update to the LI-RADS TR assessment algorithms, and they now include the presence of mass-like enhancement (any degree, any phase) in the treated lesion, along its margin, or at the surgical margin. This enhancement may appear nodular, smooth, or irregular, distorting the surrounding parenchyma.
- The detection of restricted diffusion (any degree) or mild-to-moderate hyperintensity on T2-weighted MR images in areas of uncertain mass-like enhancement can be used at the radiologist's discretion to "upcategorize" lesions from LR-TR equivocal to LR-TR viable.
- Tumor response after radiation-based treatment may not be immediate, evolving gradually through various mechanisms of cell death. Early posttreatment imaging frequently reveals persistent masslike enhancement and ancillary features suggestive of tumor viability, requiring careful evaluation of changes over time.
- After radiation-based treatment, ancillary features include new or increased restricted diffusion or mild-to-moderate hyperintensity on T2-weighted MR images, which must correspond to the area of stable or decreased masslike enhancement.

Recognizing the complexity of assessing treatment response (TR), the American College of Radiology (ACR) introduced the TR assessment algorithm as part of the Liver Imaging Reporting and Data System (LI-RADS) in 2017. This system standardizes the evaluation and reporting of individual lesions in patients treated with locoregional therapy who are at high risk for HCC. The 2017 LI-RADS TR assessment algorithm allows categorization of treated lesions as nonevaluable, nonviable, equivocal, or viable based on the probability of residual viable tumor (4).

Studies (5,6) have demonstrated that posttreatment imaging features differ substantially based on therapy type. Nonradiation-based therapies such as ablation and embolization typically produce immediate tumor destruction. In comparison, radiation-based therapies cause delayed tumor cell death, resulting in persistent imaging findings that evolve over time (7,8). These distinct response patterns revealed limitations in applying a single TR assessment approach across all therapy types. Incorporating emerging evidence and user feedback (9), the ACR updated the LI-RADS TR assessment criteria in 2024 (Fig 1). This update introduced a dedicated algorithm for radiation-treated lesions, simplified the definition of viable disease, and incorporated ancillary imaging features to upgrade lesions to the viable category (10). Lyschik et al (11) described a complementary

LI-RADS TR assessment algorithm for evaluation of lesions with contrast-enhanced US.

The following sections of this review outline the ACR's updated 2024 LI-RADS TR assessment criteria, highlighting their clinical implications and providing practical guidance for imaging interpretation after locoregional therapy and surgical resection.

CT and MRI Techniques

Multiphasic contrast-enhanced CT or MRI is essential to evaluate TR after locoregional therapy. CT involves the use of iodine-based contrast material and should also include an unenhanced sequence to distinguish true enhancement from posttreatment changes. MRI involves either extracellular or hepatobiliary contrast agents, such as gadobenate dimeglumine or gadoxetate disodium.

The minimum protocol for both CT and MRI includes imaging during the arterial and portal phases as well as during either the delayed phase or transitional and hepatobiliary phases, depending on the contrast agent used. Acquisition during the late arterial phase is critical for detecting residual tumor because this phase coincides with peak enhancement and optimal lesion-to-background contrast. This phase can be identified based on portal vein enhancement: The early arterial phase shows no portal enhancement, while the late arterial phase demonstrates at least some portal enhancement relative to that of the liver parenchyma (10,12,13).

Additional required MRI sequences include unenhanced T1-weighted (in-phase and out-of-phase) and T2-weighted imaging. Diffusion-weighted and subtraction MRI are recommended to enhance diagnostic accuracy (10). Diffusion-weighted MRI sequences generate signal intensity contrast through detection of restriction of the random motion of water molecules. After the patient undergoes locoregional therapy, restricted diffusion reflects residual hypercellularity, potentially improving detection of any residual tumor tissue. Diffusion-weighted MRI protocols should include imaging with at least two b values ($b = 0$ –50 sec/mm² and 500–1000 sec/mm²), with corresponding apparent diffusion coefficient maps to distinguish true diffusion restriction from T2-weighted "shine-through."

Subtraction MRI techniques digitally remove baseline signal intensity from postcontrast T1-weighted MR images and selectively demonstrate perfusion-related signal intensity. To obtain valid subtraction MR images, acquisition parameters between sequences must be identical. Accurate coregistration is essential because even minor positional differences generate misregistration artifacts that appear as high- or low-signal-intensity bands around structures (10,12) (Fig S1). A summary of the imaging protocol for MRI is provided in Table 1.

Nonradiation Therapy Response Assessment

Treatment Methods

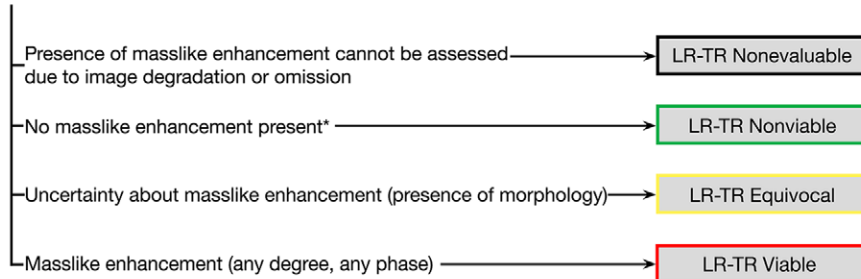
Nonradiation-based therapies include liver resection, locoregional therapy, and transcatheter embolization. These approaches achieve their effects immediately, either by inducing necrosis or by removing the tumor. Surgical resection



LI-RADS® CT/MRI Nonradiation TRA v2024 Core

Observation treated by nonradiation-based LRT or at surgical margin after resection, imaged with multiphase CT/MRI in at-risk patient

Step 1. Apply nonradiation TRA decision tree to assess for masslike enhancement (any degree, any phase), in treated lesion, along treated lesion margin, or along surgical margin

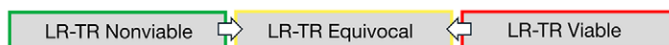


Step 2 (Optional). Apply ancillary features (AFs) favoring viability to upgrade from LR-TR Equivocal to LR-TR Viable:

- What: diffusion restriction or mild-moderate T2 hyperintensity
- Where: in area of uncertain masslike enhancement



Step 3. Apply tiebreaking rules if needed: If unsure between two TRA categories, choose category reflecting lower certainty



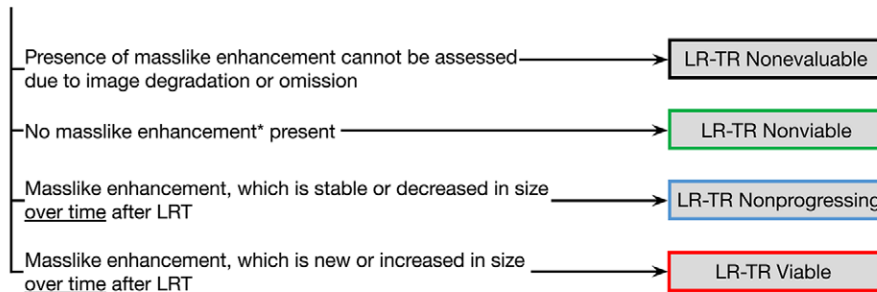
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LI-RADS® CT/MRI Radiation TRA v2024 Core

Observation treated by radiation-based LRT, images with multiphase CT/MRI in at-risk patient

Step 1. Apply radiation TRA decision tree to assess for masslike enhancement (any degree, any phase), and its change over time after LRT, in treated lesion or along its margin



Step 2 (Optional). Apply ancillary features (AFs) favoring viability to upgrade from LR-TR Nonprogressing to LR-TR Viable:

- What: diffusion restriction (any degree) or mild-moderate T2 hyperintensity, which is new or increased in size **over time** after LRT
- Where: in area of stable or decreasing masslike enhancement



Step 3. Apply tiebreaking rules if needed: If unsure between two TRA categories, choose category reflecting lower certainty



B

Figure 1. LI-RADS 2024 TR assessment (TRA) algorithms for nonradiation-based locoregional treatment (LRT) (A) and radiation-based LRT (B). (Reprinted from reference 10 under a CC BY-ND 4.0 license.)

Table 1: MRI Protocol for LI-RADS TR Assessment

MRI Technique	Acquisition Type	Fat Saturation	Section Thickness (mm)
T1-weighted out-of-phase and in-phase (unenhanced)	Dixon dual-phase acquisition	None	If 2D: ≤ 8 mm, section gap ≤ 2 mm; if 3D: ≤ 6 mm
Axial T2-weighted	2D SSFSE or 2D FSE or TSE	Optional	≤ 8 mm, section gap ≤ 2 mm
Coronal T2-weighted	2D SSFSE or 2D FSE or TSE	Optional	≤ 8 mm, section gap ≤ 2 mm
T1-weighted arterial phase sequence	3D GRE (recommended) or 2D GRE	Strongly recommended	≤ 5 mm
T1-weighted portal phase sequence	3D GRE (recommended) or 2D GRE	Strongly recommended	≤ 5 mm
T1-weighted delayed phase sequence with an extracellular contrast agent	3D GRE (recommended) or 2D GRE	Strongly recommended	≤ 5 mm
T1-weighted transitional phase sequence with gadoxetate disodium	3D GRE (recommended) or 2D GRE	Strongly recommended	≤ 5 mm
T1-weighted hepatobiliary phase sequence with gadoxetate disodium	3D GRE (recommended) or 2D GRE	Strongly recommended	≤ 5 mm
Diffusion-weighted sequence (optional)	Single-shot echo planar with more than two b values ($b = 0$ –50 sec/mm ² and 400–1000 sec/mm ²)	NA	≤ 8 mm, section gap ≤ 2 mm

Source.—Reference 10.

Note.—FSE = fast spin-echo, GRE = gradient-recalled-echo, NA = not applicable, SSFSE = single-shot fast spin-echo, TSE = turbo spin-echo, 3D = three-dimensional, 2D = two-dimensional.

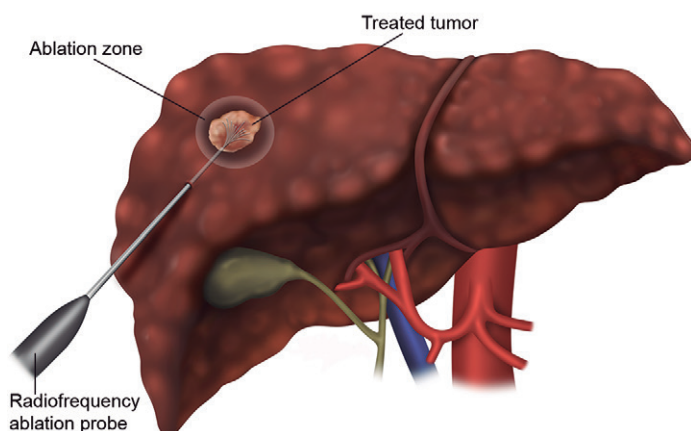


Figure 2. Illustration shows ablative therapy, a local treatment method that uses chemical or energy-based approaches to induce tumor cell death. In energy-based methods such as radiofrequency ablation and microwave ablation, applicators are inserted directly into the lesion, creating an ablation zone that encompasses the tumor and a 5–10-mm margin to address potential microscopic disease. Ablation can serve as curative treatment of early-stage disease, typically for lesions smaller than 3 cm.

is the primary curative therapy for localized HCC in patients without cirrhosis. In those with cirrhosis, resection may be considered for single tumors of any size, provided that sufficient hepatic function can be maintained after surgery. The decision for surgical intervention requires a comprehensive assessment of portal hypertension, liver function, and planned resection volume (2,3).

Locoablative therapy includes chemical ablation (percutaneous ethanol ablation) and energy-based methods, primarily radiofrequency and microwave ablation. In both energy-based methods, needle-like applicators deliver energy

as heat, reaching temperatures of 60°–100°C at the center of the applicator and propagating outward to induce coagulative necrosis (Fig 2). The resulting ablated area includes a 5–10-mm margin surrounding the lesion to address potential microscopic disease (14).

Ablative therapy can be performed with curative intent for patients with early-stage HCC. In patients with a solitary tumor smaller than 2 cm, ablation and surgical resection have comparable outcomes and are equally viable options (15). For central tumors, ablation is typically preferred because surgical resection requires major hepatectomy. Ablative methods are less effective for lesions larger than 3 cm or those adjacent to major blood vessels due to energy dissipation (2,3).

Transcatheter therapy delivers therapeutic agents to the feeding arteries of the tumor to induce necrosis (Fig 3A). Because the background liver is predominantly supplied by the portal vein, selective arterial catheterization enables targeted therapy while minimizing liver injury. Transcatheter methods include transarterial embolization, transarterial chemoembolization (TACE), and drug-eluting bead TACE (Fig 3B). Transarterial radioembolization with yttrium 90 (⁹⁰Y) follows a similar technique but primarily exerts its effect through radiation.

TACE involves infusion of a chemotherapeutic agent, typically doxorubicin, which can be mixed with ethiodized oil (ie, Lipiodol; Guerbet). The radiopaque oil enables assessment of therapy dispersion on CT images, with the extent of the accumulation of oil correlating with tumor response after TACE. Diffuse oil distribution in the treated tumor is associated with near-complete necrosis at pathologic examination, compared with absent, patchy, or partial oil coverage (16).

Transcatheter therapy is the standard option for intermediate-stage HCC, improving survival rates over those for supportive care (17). It may be considered for patients with

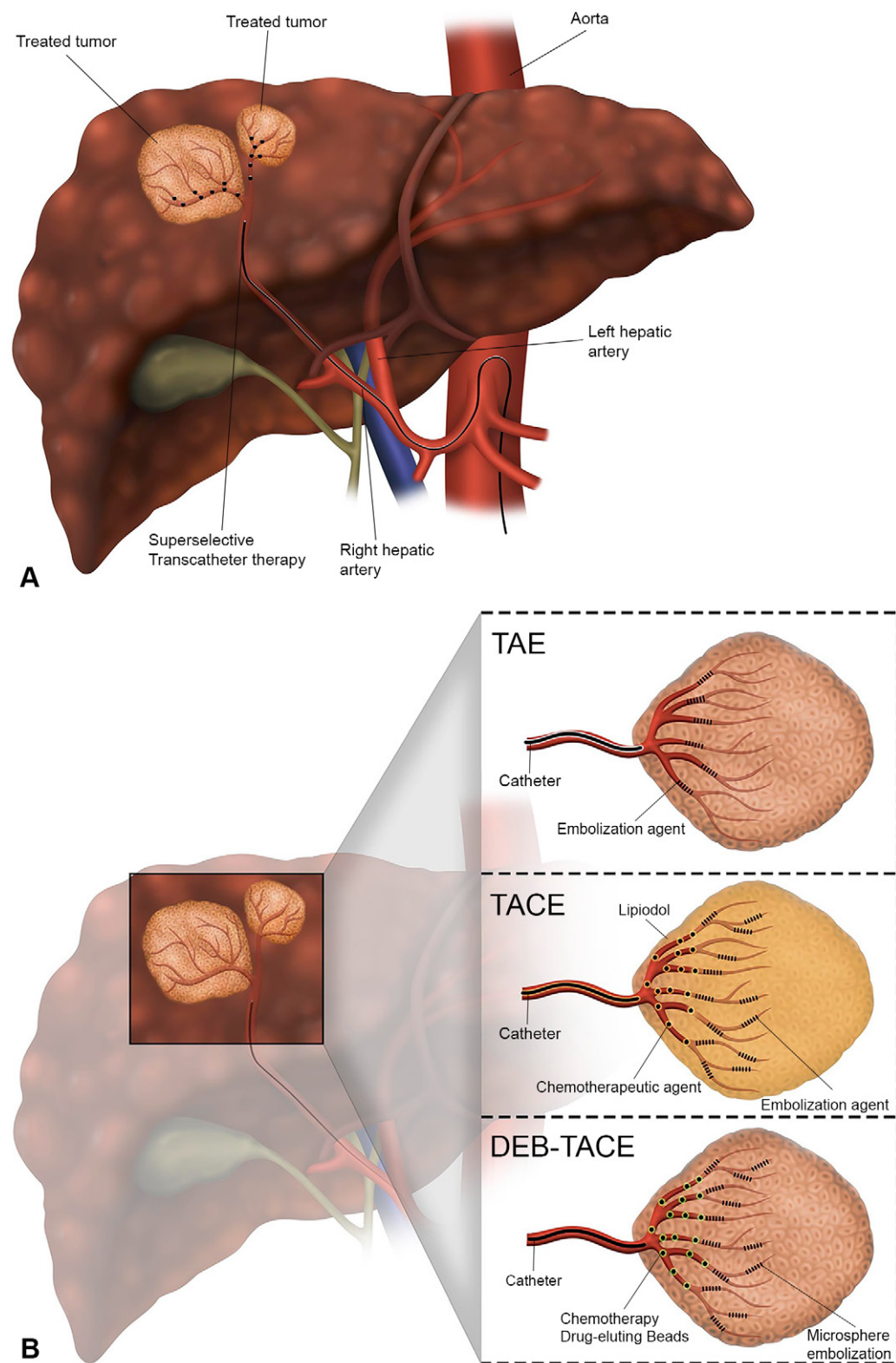


Figure 3. Illustration of nonradiation-based transcatheter therapies. **(A)** Hepatic arteries are selectively catheterized to deliver embolic material and chemotherapeutic agents, inducing ischemia and cytotoxic necrosis. These procedures serve as standard treatment of intermediate-stage HCC and can be used for early-stage disease when primary treatments are contraindicated or unfeasible. **(B)** Transarterial treatments share similar technical approaches but differ in the composition of injectable material and the mechanism of action. Transarterial embolization (TAE) interrupts the tumor's arterial supply with an embolic agent (eg, gelatin foam, gelatin particles). Transarterial chemoembolization (TACE) delivers a mixture of chemotherapeutic agents and ethiodized oil, followed by arterial embolization. Ethiodized oil serves as a carrier, allowing slow release of the chemotherapeutic agents. Drug-eluting bead TACE (DEB-TACE) uses microspheres preloaded with chemotherapy, enabling deeper embolization of smaller vessels that provides prolonged drug release.

early-stage HCC when other treatments are not feasible or are contraindicated. Relevant contraindications include portal vein thrombosis and hepatic decompensation due to the risk of additional injury from the procedure. Current guidelines (2,3) do not establish a preference among TACE, drug-eluting bead TACE, and transarterial embolization because randomized trials (18–20) have shown comparable efficacy.

TR Categories

Nonradiation-based therapy induces immediate cell death due to ischemia or direct cellular damage, which is evidenced by the disappearance of tumoral enhancement at posttreatment imaging (5,6). Therefore, the detection of residual or new masslike enhancement in the treated lesion or along its margins indicates the presence of a viable tumor.

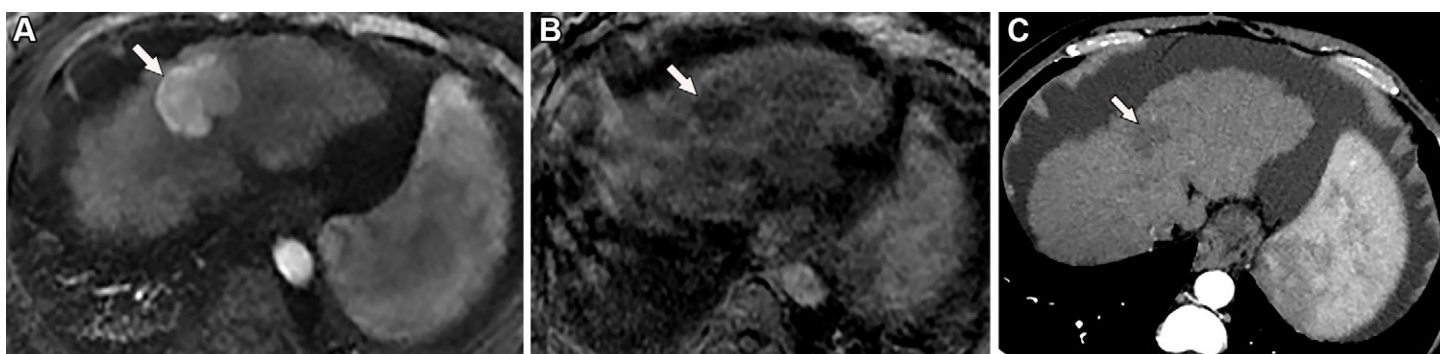


Figure 4. LR-TR nonevaluable category lesion after TACE. **(A)** Pretreatment MR image acquired during the late arterial phase shows a 45-mm lesion involving segments VIII and IVa with arterial phase hyperenhancement (arrow), categorized as LI-RADS 5 (definite HCC). **(B)** Follow-up arterial phase MR image acquired 3 months after TACE shows poor image quality due to respiratory motion artifacts and early arterial phase timing, preventing accurate assessment of masslike enhancement in the treated lesion (arrow), which was categorized as LR-TR nonevaluable. **(C)** Subsequent arterial phase CT image acquired 6 months after TACE shows the absence of lesional enhancement (arrow). The lesion was categorized as LR-TR nonviable.

LR-TR Nonviable

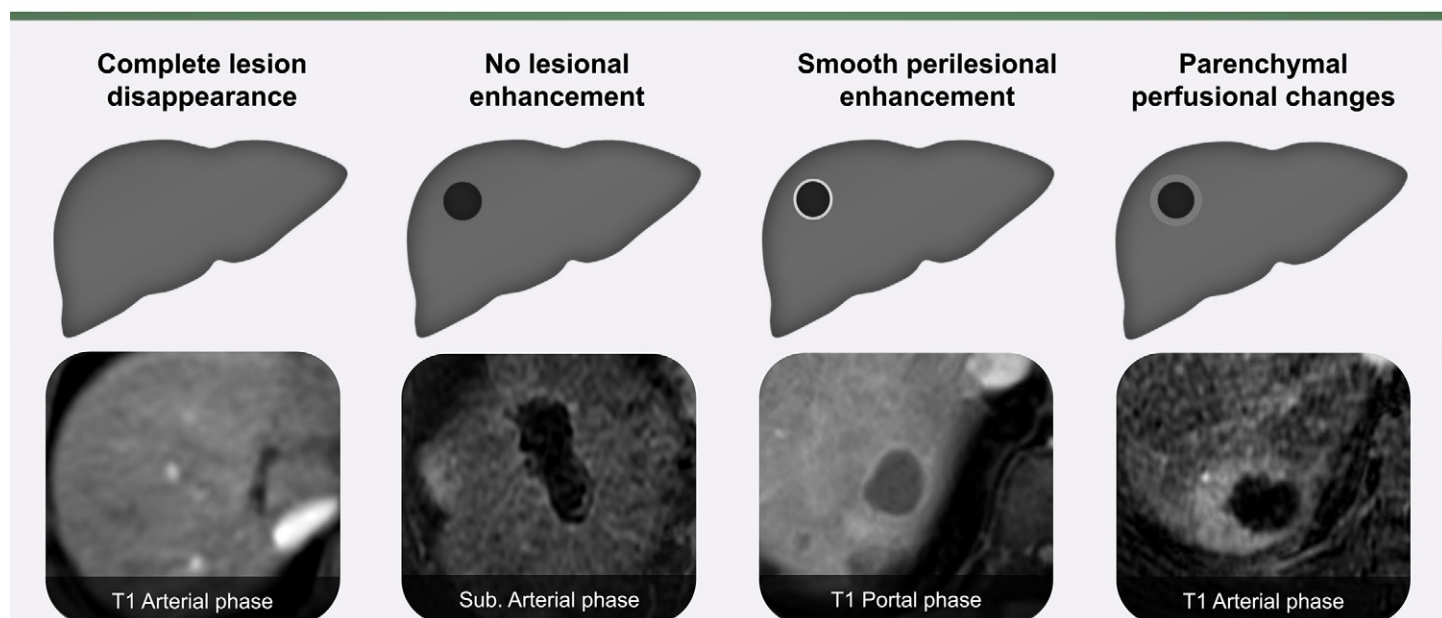


Figure 5. Illustrations and MR images show lesions with an LR-TR nonviable response. These include (L to R) complete lesion disappearance, absence of lesional enhancement, smooth perilesional enhancement, and parenchymal perfusional changes, without masslike features. Smooth perilesional enhancement appears as a thin regular rim of enhancement surrounding the treated lesion, attributable to granulation tissue. Perfusional changes are characterized by arterial phase hyperenhancement, typically demonstrating isoenhancement on MR images acquired during the portal venous and delayed phases.

LI-RADS TR Nonevaluable.—The LI-RADS (LR)-TR category is assigned when masslike enhancement cannot be evaluated due to poor image quality or technical limitations (eg, missing images from contrast-enhanced phases) (Fig 4). The underlying issue should be reported to suggest an alternative modality, if needed. When technical limitations preclude evaluation, the nonevaluable category communicates that assessment is inconclusive and alternative imaging should be performed or follow-up intervals should be shortened.

LR-TR Nonviable.—LR-TR nonviable is defined by the absence of masslike enhancement (any degree, any phase) at the treated lesion, along its margin, or at the surgical margin. Exam-

ples of findings observed in this category include complete disappearance of the lesion, no intralesional enhancement, smooth perilesional enhancement, and parenchymal perfusional changes (Fig 5).

The imaging appearance of lesions evolves after locoregional therapy, often demonstrating low signal intensity on T1-weighted and T2-weighted MR images, or high signal intensity on T1-weighted MR images in the presence of coagulative necrosis (21–23). Occasionally, an area of increased signal intensity on T2-weighted MR images may reflect liquefactive necrosis or a biloma (Fig 6). Intralesional gas may persist for up to 1 month and should not be confused with infection in asymptomatic patients (23). On contrast-enhanced MR images, a

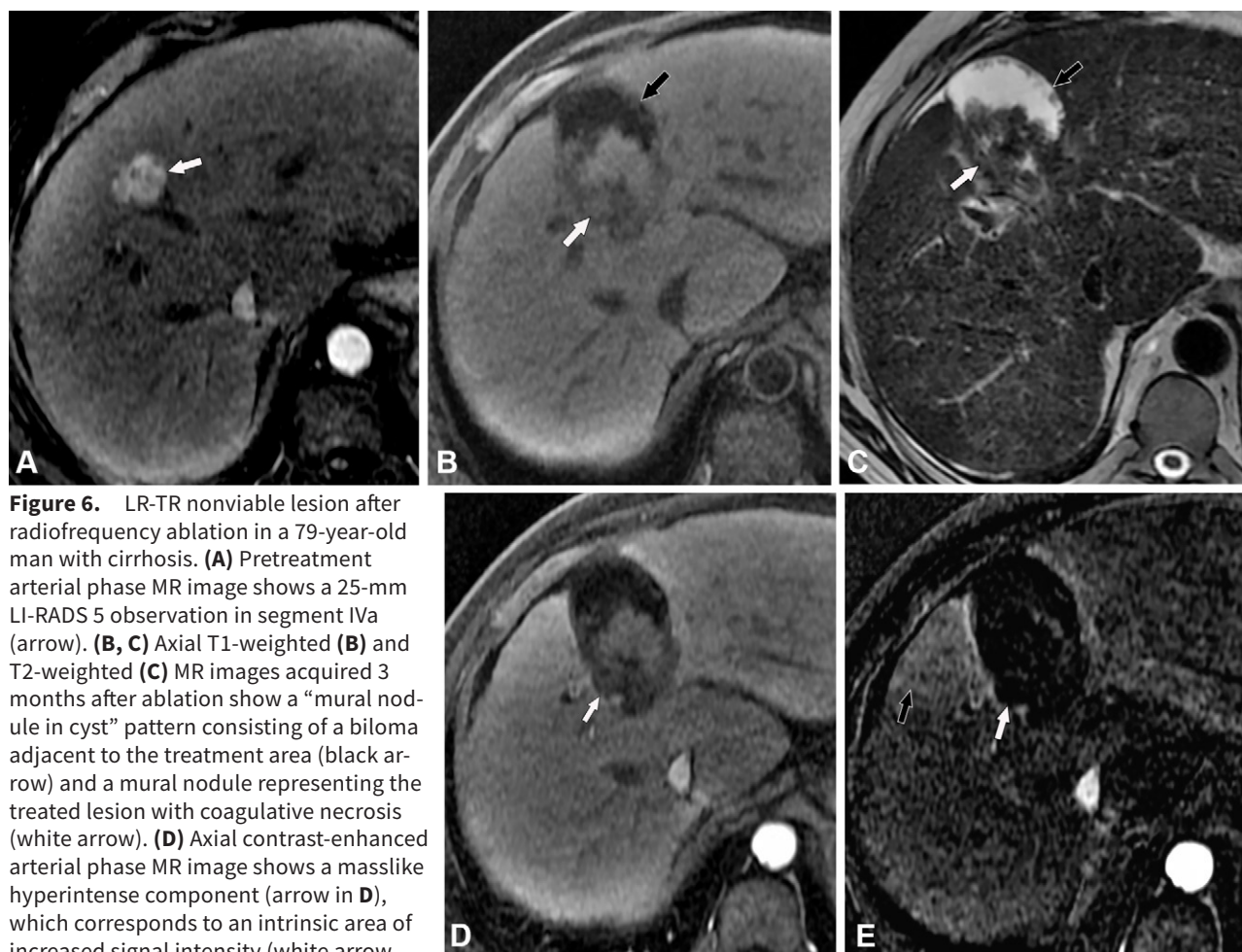


Figure 6. LR-TR nonviable lesion after radiofrequency ablation in a 79-year-old man with cirrhosis. **(A)** Pretreatment arterial phase MR image shows a 25-mm LI-RADS 5 observation in segment IVa (arrow). **(B, C)** Axial T1-weighted **(B)** and T2-weighted **(C)** MR images acquired 3 months after ablation show a “mural nodule in cyst” pattern consisting of a biloma adjacent to the treatment area (black arrow) and a mural nodule representing the treated lesion with coagulative necrosis (white arrow). **(D)** Axial contrast-enhanced arterial phase MR image shows a masslike hyperintense component (arrow in **D**), which corresponds to an intrinsic area of increased signal intensity (white arrow in **B**). **(E)** Axial subtraction arterial phase MR image shows the absence of masslike enhancement (white arrow), allowing the lesion to be categorized as LR-TR nonviable. Benign perilesional perfusional enhancement is noted (black arrow).

smooth rim of perilesional enhancement attributed to granulation tissue surrounding the treated area may be observed and can persist for months to years (24).

Perfusional changes affecting the surrounding parenchyma are commonly observed after locoregional treatment (Fig 5). Transient hyperemia is a frequent postablation finding, characterized by a rim of ill-defined arterial enhancement surrounding the treated area. Geographic arterial enhancement with isoenhancement to the parenchyma on portal and delayed phase images can also be observed, likely reflecting inflammation and arteriportal shunts (23–25). Although these perfusional alterations regress over time, they require careful evaluation in subsequent studies because treatment-related enhancement may obscure small foci of viable tumor.

LR-TR Equivocal.—The LR-TR equivocal category is assigned when there is uncertainty regarding the presence or morphology of masslike enhancement in a treated lesion, along its margin, or at the surgical margin (Fig 7). LR-TR equivocal observations harbor pathologic viability in 33%–83% of cases (26–30). Despite a substantial probability of residual viable disease, the overall survival time in patients with LR-TR equivocal observations is similar to that in those with LR-TR nonviable observations and is significantly better than in

those with LR-TR viable observations (31,32). As a result, close imaging follow-up is typically performed because unwarranted retreatment for indeterminate findings may contribute to hepatic decompensation and poor clinical outcomes (33,34).

LR-TR Viable.—LR-TR viable criteria were simplified in the 2024 update to the LI-RADS TR assessment algorithms, and they now include the presence of masslike enhancement (any degree, any phase) in the treated lesion, along its margin, or at the surgical margin. This enhancement may appear nodular, smooth, or irregular, distorting the surrounding parenchyma (Fig 8). The LI-RADS 2017 TR assessment criteria for LR-TR viable included nodular, masslike, irregular, or irregularly thick tissue, with arterial phase hyperenhancement, washout, or enhancement similar to that before the patient underwent treatment; the latter two were specific features expanding on the modified Response Evaluation Criteria in Solid Tumors (RECIST) for HCC (4).

In a meta-analysis from 2021 (35), the authors evaluated these criteria against pathologic examination results and found that arterial phase hyperenhancement had the highest sensitivity for detecting incomplete pathologic necrosis (67%), followed by washout (43%), and enhancement similar to that before treatment (24%). Among lesions with incomplete

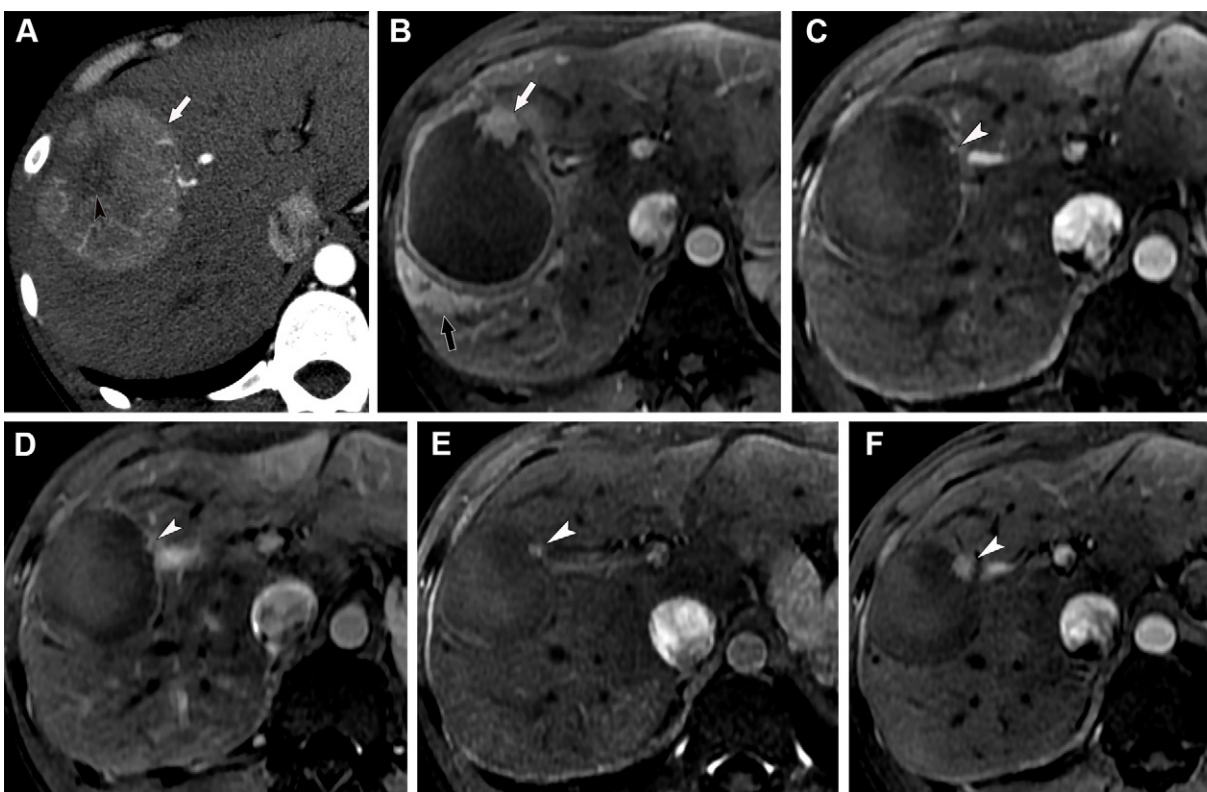


Figure 7. LR-TR equivocal lesion evolving into LR-TR viable disease after TACE in a pathologically confirmed fibrolamellar HCC in a 32-year-old man. **(A)** Axial arterial phase pretreatment CT image shows a 76-mm mass in segment VIII (arrow), with arterial phase hyperenhancement and a central scar. **(B)** Axial arterial phase MR image acquired 3 months after TACE shows residual masslike enhancement at the medial margin (white arrow), categorized as LR-TR viable. Marked perilesional enhancement represents perfusional changes (black arrow). **(C, D)** Axial arterial phase MR image at 1 month **(C)** and 3 months **(D)** after a second TACE procedure shows a subtle area of arterial phase hyperenhancement (arrowhead), measuring 3 mm at 1 month and 4 mm at 3 months. Because the enhancement was not definitely masslike, the lesion was categorized as LR-TR equivocal at both time points. **(E, F)** Axial arterial phase MR images at 9 months **(E)** and 12 months **(F)** after the second TACE show interval growth and definite masslike enhancement (arrowhead), now measuring 7 mm and 15 mm, respectively, categorized as LR-TR viable.

pathologic necrosis, only 2% exhibited washout without arterial phase hyperenhancement, while enhancement similar to that before treatment allowed detection of an additional 1% of viable tumors, nearly always with washout (35).

A separate meta-analysis (36) showed that the highest interreader agreement was for arterial phase hyperenhancement ($\kappa = 0.67$ – 0.75), compared with that for washout ($\kappa = 0.52$ – 0.64) and that for enhancement similar to pretreatment agreement ($\kappa = 0.47$ – 0.76). In conjunction, available evidence indicates that arterial phase hyperenhancement has a larger role in predicting viable tumors than do other features, warranting an adaptation that prioritizes masslike enhancement as the primary diagnostic feature.

Ancillary Features

The updated 2024 LI-RADS TR assessment criteria introduced ancillary features intended to enhance the detection of residual tumors in cases of equivocal enhancement. The detection of restricted diffusion (any degree) or mild-to-moderate hyperintensity on T2-weighted MR images in areas of uncertain masslike enhancement can be used at the radiologist's discretion to "upcategorize" lesions from LR-TR equivocal to LR-TR viable (Fig 9).

Validation studies of the 2017 TR assessment criteria have shown that LR-TR nonviable and LR-TR viable criteria demonstrate high specificity for detection of complete and incomplete pathologic necrosis, respectively (28,35). In comparison, LR-TR equivocal criteria have shown lower accuracy in assessing the extent of tumor necrosis, with radiologic-pathologic correlation studies indicating residual tumor in 33%–83% of equivocal cases (26–30). Because features of viability are more frequently observed in larger residual disease, equivocally viable components tend to be smaller than viable components (29,30).

Sensitivity for viable tumors might not be enough to guide treatment because meta-analyses (35,36) have shown it to be approximately 60%. Incorporating LR-TR equivocal and LR-TR viable criteria improves sensitivity and positive predictive value for incomplete necrosis, albeit with a reduction in specificity (29,37). A counterargument to combining LR-TR equivocal and LR-TR viable criteria is that the outcomes of the former resemble those with LR-TR nonviable criteria (31,32). However, identification of LR-TR equivocal cases where early intervention may be beneficial is an important clinical dilemma. Evolving research explores the integration of ancillary features with LR-TR equivocal criteria, specifically diffusion

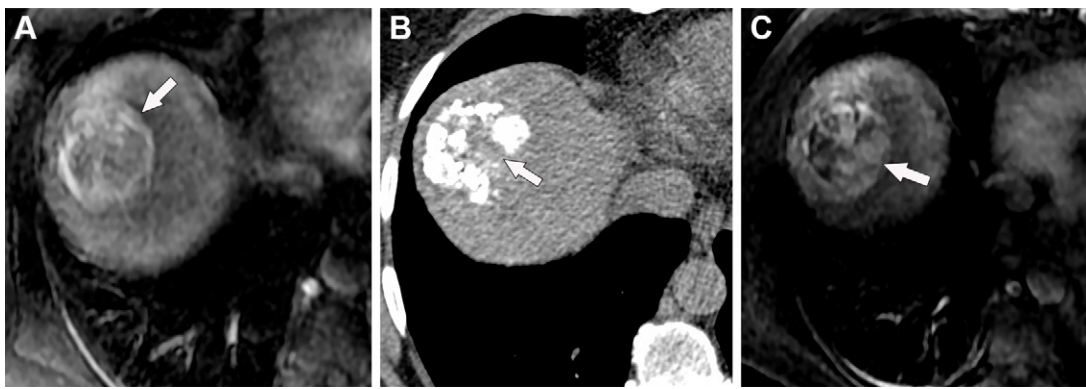


Figure 8. LR-TR viable lesion after TACE in a 62-year-old woman with cirrhosis. **(A)** Pretreatment axial arterial phase MR image shows a 55-mm LI-RADS 5 observation in segment VIII (arrow). **(B)** Axial nonenhanced CT image acquired after TACE shows partial oil coverage (arrow), a factor associated with an increased risk of incomplete tumor necrosis. **(C)** Axial arterial phase MR image at 3 months after TACE shows masslike enhancement corresponding to areas of poor oil coverage (arrow) measuring 40 mm, categorized as LR-TR viable.

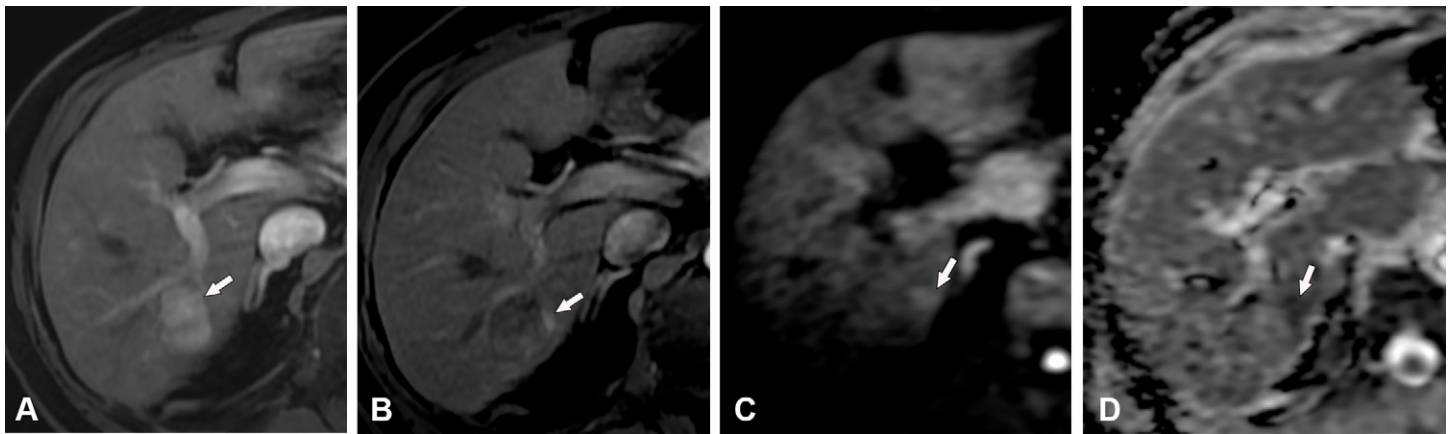


Figure 9. LR-TR viable lesion after TACE in a 69-year-old man with cirrhosis, upgraded based on ancillary imaging features. **(A)** Pretreatment axial arterial phase MR image shows a 38-mm LI-RADS 5 observation in segment VI (arrow). **(B)** Arterial phase MR image acquired 3 months after TACE shows residual heterogeneous enhancement in the treated area (arrow), categorized as equivocal for masslike enhancement. **(C, D)** Axial diffusion-weighted MR image **(C)** shows additional ancillary features, and apparent diffusion coefficient map **(D)** shows restricted diffusion (arrow), corresponding to the area of equivocal enhancement, allowing "up-categorization" to LR-TR viable.

restriction, moderate hyperintensity at T2-weighted MRI, and hypointensity at hepatobiliary phase MRI, demonstrating improved sensitivity for detecting viable tumors while maintaining specificity (28,29,38,39). Despite evidence supporting the use of hepatobiliary hypointensity, it was not included in the 2024 TR assessment criteria due to concerns regarding general applicability and diagnostic reliability (40,41).

Further supporting these refinements, a validation study of the 2024 nonradiation TR assessment criteria incorporating ancillary features demonstrated significantly improved sensitivity and accuracy for detecting incomplete necrosis, when compared with 2017 criteria (42). The slight decrease in specificity found in this study underscored the need to ensure that the area with ancillary features is colocalized to the area with equivocal enhancement.

Postsurgical Findings

Nonradiation TR assessment criteria can be applied to assess residual tumor at the surgical margin for hepatectomy and segmentectomy or at the surgical cavity for smaller (ie,

wedge) resections. This cavity can be filled with fluid or fat after surgery but should not display masslike enhancement. Recurrence elsewhere in the liver should be evaluated using the diagnostic algorithm.

Common postoperative findings include fluid collections along the resection margin or within subphrenic and perihepatic spaces, which may represent hematomas, bilomas, or seromas. The surgical margin may resemble adjacent liver parenchyma or display low attenuation at CT and mild hyperintensity at T2-weighted MRI, reflecting transient accumulation of blood and bile (43,44). Edema and regenerative changes along the resection margin can appear as irregular, wedge-shaped, or serpiginous regions with mild-to-moderate arterial enhancement (Fig S2).

Radiation TR Assessment

Radiation Damage Mechanism

Radiation therapy aims to inhibit cellular replication and induce cell death by causing substantial DNA damage. This

damage results from direct ionization of DNA molecules or through indirect mechanisms involving the production of reactive oxygen species (45,46).

Radiation-induced damage activates cellular signaling pathways that regulate the modality of cell death, including (a) apoptosis, a gene-regulated active form of cell death with rapid onset; (b) necrosis, an uncontrolled and inflammatory cell death, potentially secondary to extensive apoptosis overwhelming the phagocytes; (c) senescence, in which cells lose their proliferative capacity but remain metabolically active; and (d) mitotic catastrophe, characterized by inadequate progression through the division phase with under-repaired DNA, resulting in cells with chromosomal aberrations (45–48).

Which of these cell death pathways predominates depends on the radiation dose and tumor characteristics, including the cell cycle phase, microenvironment, and tissue of origin. Immediate cell death through apoptosis or necrosis may occur in highly radiosensitive tumors or at higher ablative doses. In solid tumors, mitotic catastrophe is the predominant response to clinical radiation therapy, often leading to delayed cell death over weeks to months (48). During this period, affected cells remain metabolically active while in cell cycle arrest, contributing to persistent imaging findings.

Treatment Methods

Historically, the role of radiation therapy in the treatment of HCC was limited due to hepatic radiosensitivity and the associated risk of radiation-induced liver disease. The development of more efficient delivery methods, such as stereotactic body radiation therapy (SBRT), along with selective approaches for transarterial radioembolization in the form of radiation segmentectomy, has enabled high tumoral doses while minimizing radiation exposure in the surrounding tissues (49,50).

SBRT uses 3D radiation planning to deliver ablative doses of radiation to the targeted tumor (Fig 10). With imaging guidance, the clinical target volume is defined by extending the gross tumor volume with a 5–10-mm margin to account for possible subclinical disease. The planned target volume is then expanded to accommodate variations in size and position throughout different respiratory phases. Treatment is delivered through multiple precise beams from diverse angles to create steep dose gradients in the target volume. Despite the rapid dose decrease, surrounding tissues receive intermediate to low doses of radiation, which limits its use for tumors near gastrointestinal structures or in patients with liver dysfunction (3,49).

Transarterial radioembolization involves targeted delivery of microspheres loaded with radioactive ^{90}Y through selective catheterization of tumor-feeding hepatic arteries (Fig 11). The microspheres lodge in the distal circulation, providing localized radiation within a 1–10-mm penetration depth. Treatment planning involves angiography with cone-beam CT, and often technetium 99m ($^{99\text{m}}\text{Tc}$) SPECT/CT to assess the lung shunt fraction and the regional dose distribution (51,52).

The treatment strategy varies based on the tumor burden and the hepatic reserve. For extensive disease, lobar or sequential bilobar approaches aim to balance effective tumor-

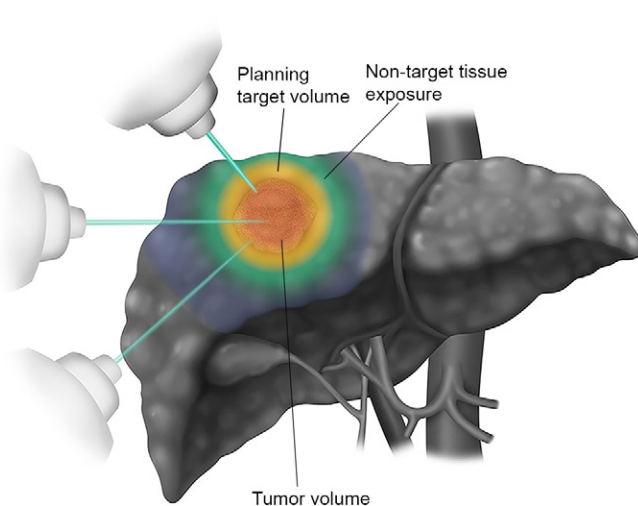


Figure 10. Illustration shows the principles of SBRT. The tumor is targeted using 3D radiation planning to deliver ablative radiation doses under imaging guidance. The clinical target volume is defined by expanding the tumor volume by a 5–10-mm margin. The planning target volume is then further expanded to account for size variations during different respiratory phases. Despite rapid dose decreases, some peritumoral parenchyma receives intermediate to low radiation doses. SBRT may be used as curative treatment of single tumors not amenable to ablative therapy or as a debulking approach to reduce tumor burden.

al doses with acceptable radiation exposure to background liver tissue. For tumors confined to a single lobe, radiation lobectomy targets that area to control growth while stimulating compensatory hypertrophy in the untreated lobe. For more limited disease, radiation segmentectomy involves the administration of high doses of ^{90}Y to superselective arteries supplying one or two adjacent segments (50,51).

Personalized dosimetry has enhanced radioembolization strategies by enabling much higher doses based on the distribution of microspheres, the volume of the tumor, and exposure of uninvolved tissues. When applied to segmental or subsegmental infusions, ablative dosages exceeding 400 Gy can be delivered provided that the dose to the background liver and the lung shunt are within safe limits (51,52). This approach has significantly improved tumor response and overall survival compared with those of conventional dosimetry methods (2,52–55). Ablative doses often achieve rapid tumor necrosis, resulting in earlier resolution of tumoral enhancement and reducing diagnostic uncertainty during follow-up.

The development of SBRT and radiation segmentectomy has allowed earlier application of radiation-based methods in the treatment paradigm. Both strategies can serve as curative alternatives to thermal ablation for patients with unresectable early-stage HCC, particularly for tumors with a higher risk of recurrence due to size or location (2,3). The LEGACY study (54) demonstrated effective tumor control with radiation segmentectomy for tumors measuring up to 8 cm, while a meta-analysis from 2019 (56) reported favorable 3-year local control outcomes for SBRT in tumors measuring up to 5 cm. In addition, SBRT and transarterial radioembolization may be considered for bridging and debulking in patients with early

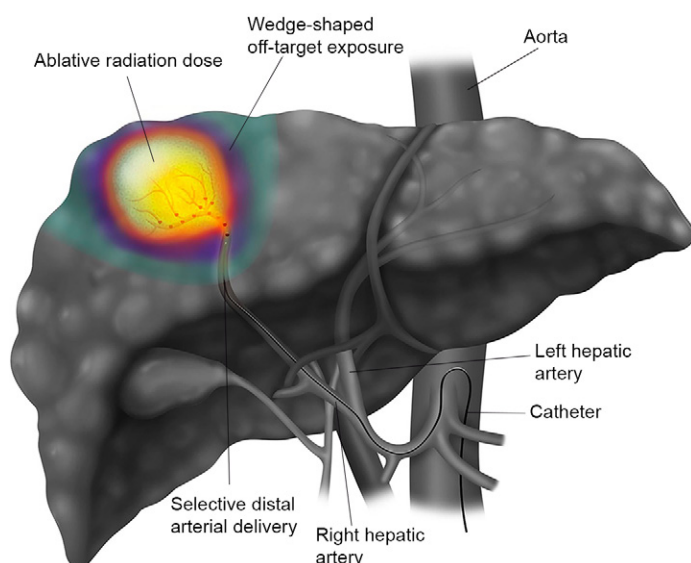


Figure 11. Illustration shows the principles of radiation segmentectomy, a procedure in which an ablative dose of ^{90}Y is selectively delivered to one or two adjacent Couinaud segments, inducing tumor apoptosis and necrosis. This procedure can serve as curative treatment of single tumors up to 8 cm, as bridging therapy before transplant, or as a method to reduce tumor burden. Off-target parenchymal radiation exposure typically demonstrates a wedge-shaped vascular pattern that follows the treated vascular territory.

to intermediate disease as well as for palliative treatment in those with advanced cases (2,51).

TR Categories

Tumor response after radiation-based treatment may not be immediate, evolving gradually through various mechanisms of cell death. Early posttreatment imaging frequently reveals persistent masslike enhancement and ancillary features suggestive of tumor viability, requiring careful evaluation of changes over time (7,8,54,57–59).

Tumors treated with radiation typically undergo a gradual reduction in size, with the greatest decrease observed within months after therapy (7,8). In most cases, masslike arterial phase hyperenhancement persists during the initial 3–6 months with stable or decreasing intensity before gradually resolving (7,8,60). Some HCCs with confirmed complete necrosis can continue to exhibit stable or slowly diminishing enhancement over a prolonged follow-up period (57,58). In comparison, increases in lesion size or new masslike enhancement indicate tumor viability (7,8). Taken together, these findings support using enhancement and changes in size of the treated tumor as imaging biomarkers to evaluate TR, forming the basis for the new categorization.

LR-TR Nonevaluable.—This category is used when TR cannot be evaluated due to suboptimal image quality or technical issues such as imaging examinations missing images from contrast-enhanced phases or pronounced motion artifacts.

LR-TR Nonviable.—LR-TR nonviable is characterized by the absence of masslike enhancement (any degree, any phase) in

the treated lesion or along its margin. Representative examples include complete disappearance of the lesion, no intralesional enhancement, smooth perilesional enhancement, and perfusional alterations, without masslike features. Radiation-induced parenchymal enhancement is an expected perfusional finding after treatment and should not be misinterpreted as evidence of a nonprogressing or viable tumor. This category is often seen immediately after radiation segmentectomy due to rapid cell death from ablative doses, or may evolve over a longer period after SBRT (7,8,61).

LR-TR Nonprogressing.—This category is assigned when treated lesions continue to exhibit masslike enhancement (any degree, any phase) within or along their margins that remains stable or decreases in size over time after radiation-based locoregional therapy (Fig 12). The LR-TR nonprogressing category replaces LR-TR equivocal in the radiation TR assessment criteria to more accurately reflect the expected appearance of treated tumors over time. This category remains applicable for as long as the treated tumor continues to exhibit residual masslike enhancement, provided that the enhancing area remains unchanged or is decreasing.

Recent studies (61,62) have demonstrated the clinical effect of the separate radiation TR assessment algorithm. The updated algorithm resulted in substantial redistribution of categories compared with previous criteria, with 57.5% of lesions that would have been categorized as LR-TR viable now categorized as LR-TR nonprogressing (62). Furthermore, long-term follow-up results (61) validated that nonprogressing lesions typically evolve to nonviability over time. This recategorization has prognostic importance because patients with LR-TR viable lesions demonstrated worse overall survival than those with LR-TR nonprogressing lesions (62).

LR-TR Viable.—The LR-TR viable category is assigned when there is a masslike enhancement (any degree, any phase) in the treated lesion that exhibits an increase in size over time or a new lesion along its margins after radiation-based treatment (Fig 13).

Ancillary Features

At the radiologist's discretion, ancillary features can optionally be applied to adjust the category LR-TR nonprogressing to LR-TR viable. After radiation-based treatment, ancillary features include new or increased restricted diffusion or mild-to-moderate hyperintensity on T2-weighted MR images, which must correspond to the area of stable or decreased masslike enhancement. Because most tumors continue to display masslike enhancement after sustaining radiation damage, assessment of viability involves observing changes throughout serial imaging examinations. However, minor differences in size may reflect measurement variability rather than actual tumor progression (63,64). In cases of unclear progression, assigning the LR-TR category is appropriate to indicate the need for continued follow-up. As an alternative, ancillary features can be used to assess viability independent of size, thus enabling radiologists to identify subtle progression and enable earlier diagnosis (Fig S3) (7,65, 66).

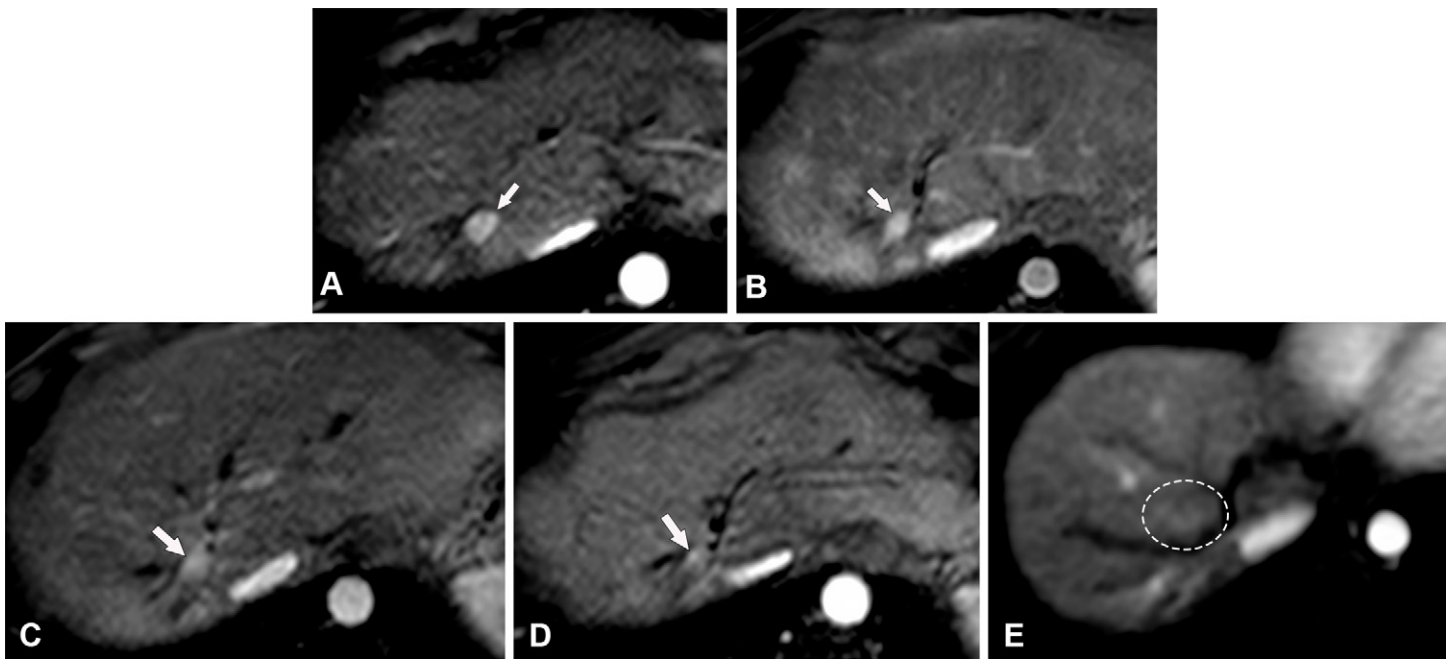


Figure 12. LR-TR nonprogressing lesion evolving to LR-TR nonviable after SBRT in a 31-year-old woman with cirrhosis of an unknown cause. **(A)** Baseline arterial phase MR image shows a 14-mm lesion in segment VII with arterial phase hyperenhancement (arrow) and hepatobiliary hypointensity (not shown), categorized as LI-RADS 4 (ie, probable HCC). Results of pathologic examination confirmed the diagnosis, and the lesion was treated with SBRT. **(B–D)** Axial arterial phase MR images after SBRT show progressive regression of the masslike enhancement (arrow) to 9 mm at 2 months after SBRT **(B)**, to 7 mm at 5 months **(C)**, and to 5 mm at 12 months **(D)**. The lesion was categorized as LR-TR nonprogressing at each time point. **(E)** Axial arterial phase MR image at 36-month follow-up shows complete disappearance of the lesion (dotted circle), categorized as LR-TR nonviable.

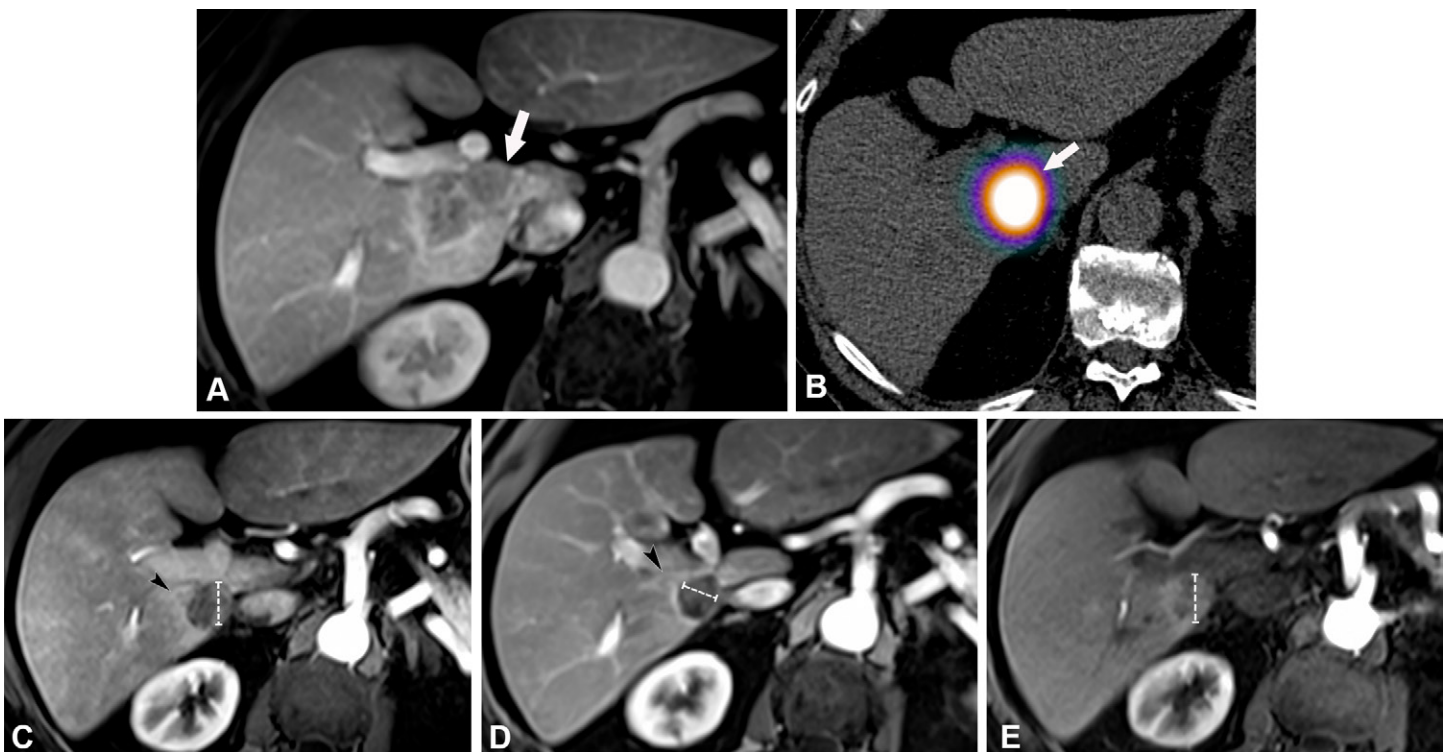


Figure 13. LR-TR viable category after radiation segmentectomy in a 76-year-old man with cirrhosis related to metabolic dysfunction–associated steatotic liver disease. **(A)** Baseline axial arterial phase MR image shows a mass with arterial phase hyperenhancement in segments V and VI measuring 50 mm (LI-RADS 5) (arrow). **(B)** Fused axial SPECT/CT image after transarterial radioembolization shows selective deposition of radioparticles in the tumor (arrow). **(C, D)** Axial arterial phase MR images show masslike enhancement measuring 18 mm (calipers) at 3 months **(C)** and 15 mm (calipers) at 6 months **(D)**. The lesion was categorized as LR-TR nonprogressing at both points. Surrounding perfusional changes due to radiation-induced sinusoidal obstruction syndrome are noted throughout (arrowhead). **(E)** Axial arterial phase MR image at 9 months reveals increased signal intensity and size of the masslike enhancement in the lateral and medial walls of the treated lesion, now measuring 17 mm (calipers) and categorized as LR-TR viable. Systemic therapy was initiated.

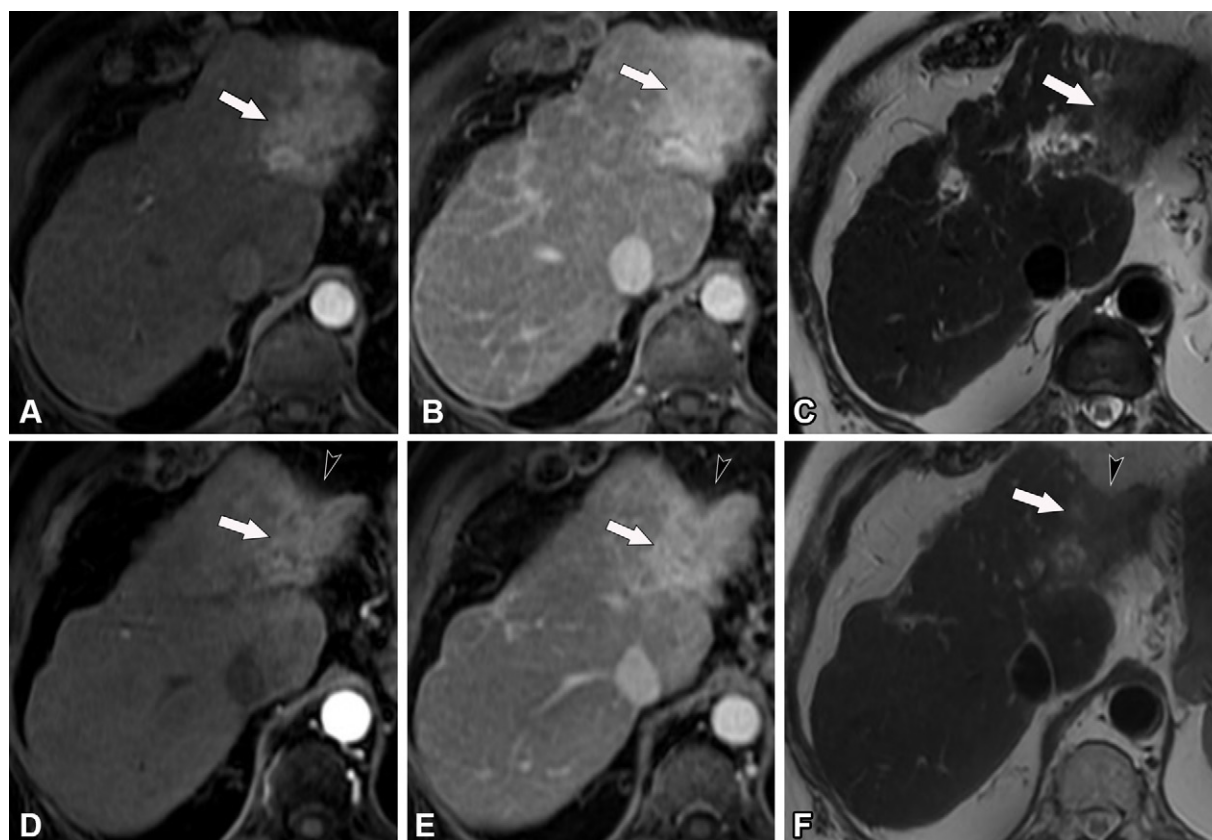


Figure 14. Radiation-induced parenchymal changes after transarterial radioembolization in a patient with cirrhosis, demonstrating evolution from acute to chronic phase injury. (A–C) Axial arterial phase (A), delayed phase (B), and T2-weighted (C) MR images at 3 months after treatment show complete lesion disappearance, categorized as LR-TR non-viable. Parenchymal changes in segment II are related to acute-phase radiation-induced sinusoidal obstruction syndrome seen as hyperenhancement (arrow in A), persistent enhancement (arrow in B), and increased signal intensity from edema (arrow in C). (D–F) Axial arterial phase (D), delayed phase (E), and T2-weighted MR images at 24 months show chronic radiation-induced parenchymal changes (arrow) seen as reduced hyperenhancement in the arterial phase (arrow in D) with increased enhancement in the delayed phase (arrow in E) and diminished T2 hyperintensity (arrow in F). Associated volumetric reduction and capsular retraction are noted (arrowhead in D–F).

Radiation-induced Parenchymal Changes

Radiation-based therapies allow delivery of most of the radiation dose to the targeted tumor, but some off-target exposure to the surrounding parenchyma is inevitable. The extent of the parenchyma affected varies according to the treatment modality, with SBRT causing peritumoral changes based on the target volume, while radiation segmentectomy produces a wedge-shaped vascular pattern (67,68). Parenchymal changes arise from radiation-induced apoptosis of endothelial cells, leading to sinusoidal obstruction syndrome (69,70).

In the acute phase (1–3 months after treatment), accumulated cellular debris, red blood cells, and fibrin obstruct sinusoidal flow, causing congestion and central venule obstruction. Disrupted blood flow leads to ischemia and necrosis of centrilobular hepatocytes.

These pathologic changes manifest at imaging as geographic areas of hypoattenuation on precontrast CT images and increased signal intensity on both T2-weighted and diffusion-weighted MR images (71,72). The increased signal intensity on diffusion-weighted MR images, with corresponding apparent diffusion coefficient hyperintensity, indicates T2-weighted “shine-through” rather than true dif-

fusion restriction (Fig S4). Postcontrast MRI shows compensatory arterial phase hyperenhancement that may persist or fade during the portal and delayed phases (67,69–71) (Fig 14A–14C). Relative portal hypoenhancement may occur due to impaired flow of contrast material from increased sinusoidal pressure. In addition, hepatocyte damage from ischemia reduces the uptake of hepatobiliary contrast material, manifesting as hypointensity on hepatobiliary phase MR images (47,67,69) (Fig S5).

During the subacute (3–6 months) and chronic (>6 months) phases, reparative processes lead to progressive fibrosis of the irradiated parenchyma. Collagen deposition by activated myofibroblasts in the perisinusoidal and perivenular spaces leads to increasing centrilobular obstruction and eventual replacement by fibrous tissue.

These changes are reflected at imaging as a gradual reduction in arterial phase hyperenhancement, while delayed phase enhancement progressively increases due to retention of contrast material in fibrotic tissue (68) (Fig 14D–14F). The affected parenchyma demonstrates a progressive reduction in signal intensity on both T2-weighted and diffusion-weighted MR images, with T2-weighted images in some patients displaying

signal hypointensity from hemosiderin deposition secondary to venous congestion (47). Concurrently, the irradiated tissue undergoes atrophy, frequently displaying architectural distortion and capsular retraction (71,72).

Lesion Measurement

For treated tumors categorized as LR-TR viable, LR-TR equivocal, or LR-TR nonprogressing, the lesion size should be documented in millimeters in the radiologic report to facilitate appropriate clinical decision making. Size measurement should represent the longest dimension of masslike enhancement in or along the lesion margin, excluding intervening nonenhancing portions and benign findings such as rim enhancement (10) (Fig 15).

Lesions can be measured in any standard orthogonal imaging plane and during any dynamic imaging phase that clearly delineates the lesion. However, the axial plane is recommended to ensure consistency in subsequent follow-up examinations. In challenging cases such as HCC with a diffuse growth pattern, the margins on postcontrast MR images can be obscured by perfusional changes or poorly discerned due to isoenhancement with the background liver. For these lesions, correlation with images from other sequences such as unenhanced T1-weighted or T2-weighted MRI improves tumor delineation for accurate measurement (73) (Fig 16).

Combined TR

Combination strategies are increasingly used in HCC treatment to enhance tumor control and address the limitations of single-modality treatments. Recent guidelines recommend additional locoregional therapy for viable disease after primary intervention, most commonly TACE, transarterial radioembolization, or SBRT (2,3).

These combination strategies require careful evaluation during follow-up assessment. Appropriate TR assessment categorization depends on the lesion's baseline features and whether subsequent treatment targeted the entire lesion or only the residual tumor. This is particularly relevant when both radiation-based and nonradiation-based treatments are involved, as accurate assessment relies on reviewing the therapies administered, their sequence, and imaging timing (74).

When radiation-based therapy is applied after nonradiation therapy, imaging interpretation should follow the radiation TR assessment algorithm. Persistent masslike enhancement within the treated residual tumor should be categorized as LR-TR nonprogressing if it is stable or decreasing in size over time. Any enlargement or new areas of masslike enhancement during follow-up indicate viable disease and should be categorized as LR-TR viable (Fig 17).

Conversely, when nonradiation treatment follows radiation-based therapy, imaging criteria for predicting viability should be based on the nonradiation TR assessment algorithm. These methods can target the entire lesion or be focused on only the portion displaying residual tumor. The presence of masslike enhancement within the newly treated area or at its margin indicates residual viable tumor and should be categorized as LR-TR viable. When treatment is selective, the remaining portion of the lesion continues to un-

Treatment Response Measurements

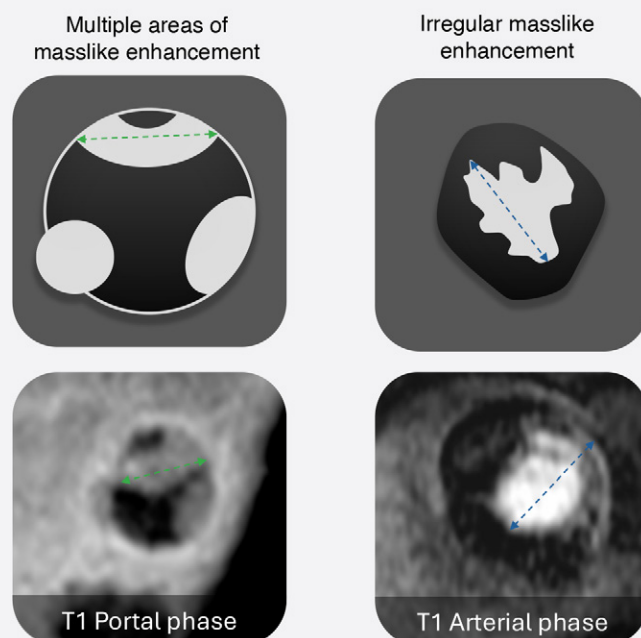


Figure 15. Illustration shows TR measurement. Treated lesions categorized as LR-TR viable, LR-TR equivocal, or LR-TR nonprogressing should be reported using a single dimension measurement of the area of masslike enhancement, excluding intervening nonenhancing areas or benign features. If multiple viable areas are present, the largest masslike enhancement should be measured.

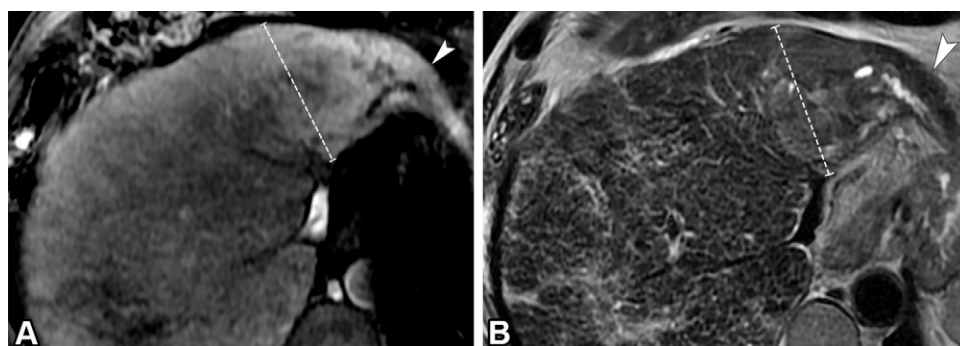
dergo delayed radiation-induced cell death. In this setting, if the newly treated area shows no viability and the remaining portion demonstrates stable or decreasing masslike enhancement, it should be classified as LR-TR nonprogressing (Fig 18).

Systemic Therapy

Systemic therapy is typically reserved for patients with advanced or intermediate-stage disease who either are ineligible for locoregional therapy or have progressed despite prior interventions. Available systemic therapies include immunotherapy, which modulates the antitumor immune response, and targeted therapies, which inhibit tumor growth pathways. First-line options include sorafenib or lenvatinib (anti-vascular endothelial growth factor [VEGF] therapy), tremelimumab plus durvalumab (checkpoint inhibitors), and atezolizumab plus bevacizumab (combined immunotherapy and anti-VEGF therapy) (2,3).

While systemic therapies are an essential part of HCC management, assessing their response presents specific challenges. The LI-RADS TR assessment criteria do not address systemic therapy alone. However, when systemic and locoregional therapies are used in combination, the algorithm may be applied with caution if locoregional treatment is the predominant contribution. Response assessment in clinical trials of systemic therapies for HCC, including pivotal phase III studies, typically relied on conventional Response Evaluation Criteria in Solid Tumors (RECIST) and modified RECIST (75–77).

Figure 16. Pathologically proven HCC, with a diffuse growth pattern, in a 50-year-old man with alcohol-related cirrhosis. **(A)** Axial arterial phase MR image shows heterogeneous hyperenhancement (arrowhead) throughout segments II and III due to tumoral tissue and perfusional changes. The tumor extends into segmental branches of the left portal vein, meeting criteria for LI-RADS definite tumor in vein (LR-TIV). **(B)** Axial T2-weighted MR image shows corresponding mild to moderate hyperintensity throughout the same segments. Diffuse HCC can be challenging to assess due



to its ill-defined margins and variable enhancement patterns. Tumor boundaries appear more conspicuous on T2-weighted images than on arterial phase MR images, where the preserved hepatic parenchyma (arrowhead) appears indistinguishable from tumor enhancement. Correlating the areas of abnormal enhancement with T2-weighted MR images better defines the tumor margins for accurate measurement (calipers in **A** and **B**). This lesion was subsequently treated with SBRT. Serial posttreatment images are presented in Figure S3.

Combined treatment: Nonradiation and Radiation therapy

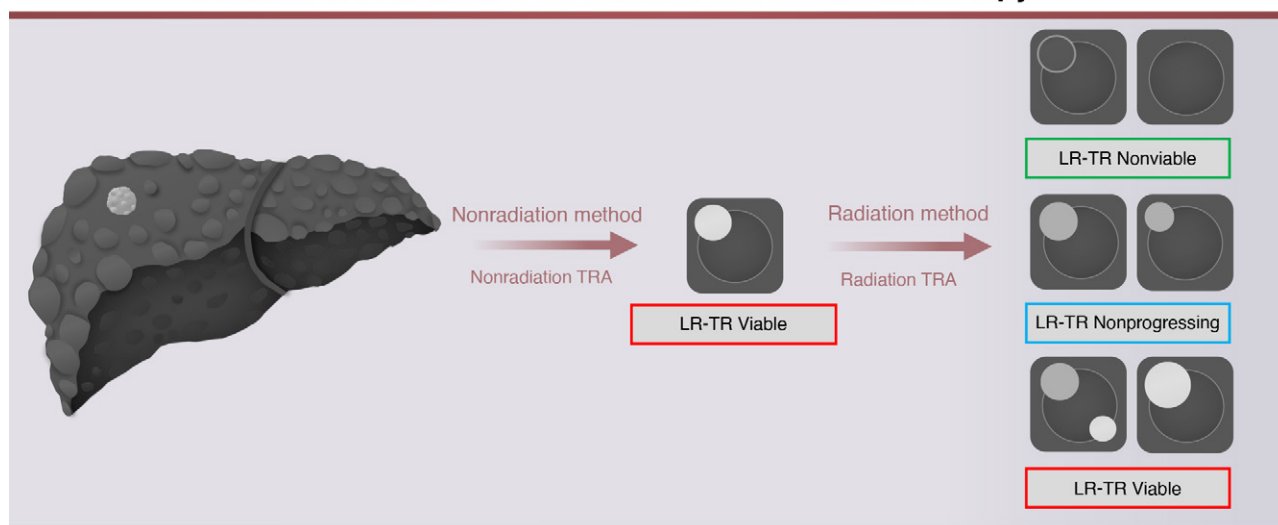


Figure 17. Illustration shows combined TR, with initial nonradiation-based therapy followed by radiation-based methods. Imaging findings should be interpreted using the radiation TR assessment algorithm. The absence of masslike enhancement in the treated lesion should be categorized as LR-TR nonviable. Persistent masslike enhancement that is stable or decreasing in size should be categorized as LR-TR nonprogressing, whereas an increase in size or the appearance of new masslike enhancement should be categorized as LR-TR viable.

Pitfalls and How to Avoid Them

Evaluation of TR can be challenging as tumoral changes, parenchymal alterations, and technical limitations may impede identification of viable disease. Misinterpretation of imaging findings can substantially affect patient care through inappropriate treatment decisions or delayed interventions. Radiologists must recognize common imaging pitfalls in posttreatment surveillance to ensure accurate assessment of TR. Table 2 summarizes the key imaging pitfalls of the LI-RADS TR assessment algorithm, why they occur, and how to avoid them.

Detection of masslike enhancement on dynamic images is the primary goal in TR assessment. Most HCCs develop high arterial flow due to angiogenesis and demonstrate peak enhancement during the late arterial phase, with some lesions displaying enhancement exclusively during this window (10).

Suboptimal acquisition or lack of acquisition during the early arterial phase may diminish the conspicuity of arterial phase hyperenhancement, limiting detection of residual disease and appropriate characterization of untreated HCC during TR assessment (10,12,13) (Fig 19).

Treatment-induced perfusional alterations can simulate enhancement related to viable tumor, potentially resulting in misdiagnosis and inappropriate retreatment. These perfusional alterations appear as wedge-shaped and geographic areas of arterial phase hyperenhancement that diminish on portal venous and delayed phase images (20–22,69,78). Additional features include isointensity or occasionally mild hyperintensity on T2-weighted MR images due to edema. Areas of perfusional enhancement lack masslike features characteristic of viable tumors, showing preserved background architecture, with vessels coursing through the region unperturbed.

Combined treatment: Radiation and Nonradiation therapy

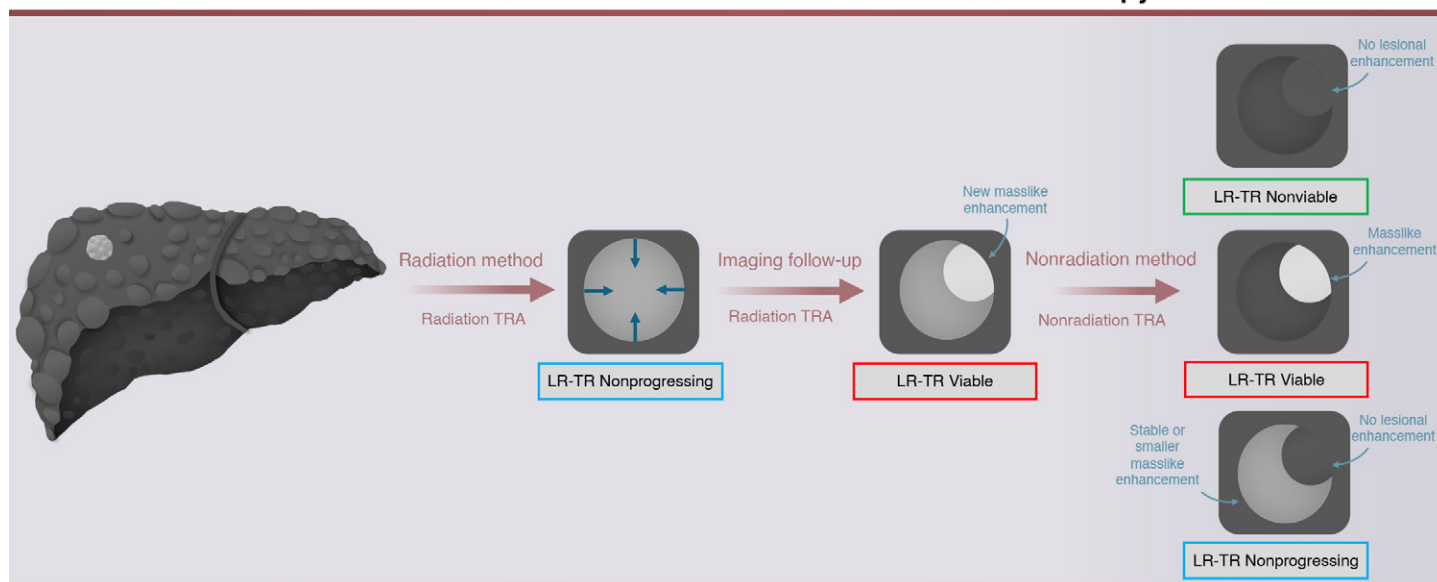


Figure 18. Illustration shows combined TR, with initial radiation-based therapy followed by nonradiation-based methods. Imaging findings should be assessed using the nonradiation TR assessment algorithm. The absence of masslike enhancement in the treated lesion should be categorized as LR-TR nonviable, while the presence of masslike enhancement in or at the margins of the treated area should be categorized as LR-TR viable. If nonradiation treatment is applied only to the viable portion, the untreated portion continues to undergo radiation-induced cell death and should be categorized as LR-TR nonprogressing.

In uncertain cases, LR-TR equivocal or LR-TR nonprogressing categorization should be assigned to prevent overdiagnosis, as this phenomenon predominates early after treatment and resolves over time.

Coagulative necrosis of the treated tumor commonly results in intrinsic hyperintensity on T1-weighted MR images that can confound evaluation of viable tumor. This appearance compromises TR assessment by simulating enhancement on postcontrast MR images or masking true enhancement by reducing contrast with the surrounding tissue (12,13,25) (Fig S6). Subtraction MRI can be used to address this issue to distinguish true enhancement from intrinsic hyperintensity, preventing misinterpretation (13,21).

Retained ethiodized oil after TACE increases parenchymal attenuation and reduces contrast with enhancing tissue, limiting detection of residual tumor tissue (25,79). Comparison between unenhanced and postcontrast images is essential to differentiate true enhancement from retained oil (Fig S7). When dense oil retention and beam-hardening artifacts preclude assessment, the LR-TR nonevaluable category should be assigned to indicate the need for use of an alternative imaging method. MRI is the preferred alternative modality because ethiodized oil does not affect fat-suppressed T1-weighted postcontrast MR images.

A frequent error in applying ancillary features is misinterpreting isolated restricted diffusion or mild to moderate hyperintensity on T2-weighted MR images as evidence of viability. Relying solely on these features can result in overdiagnosis of viable tumor, given that complete necrosis can maintain these characteristics in 8%–10% and 13%–43% of cases, respectively (29,38,39). To maintain specificity when detecting viable tumor, ancillary features must be precisely

colocalized to the area of equivocal enhancement after nonradiation-based treatments and to stable or decreasing masslike enhancement after radiation-based treatments.

When evaluating diffusion restriction, radiologists should recognize benign causes of this finding that mimic residual disease. Intralesional hemorrhage and liquefactive necrosis in the treated area can produce diffusion restriction in the absence of viable tumor (Fig S6D, S6E). T2-weighted shine-through represents another source of false-positive diffusion restriction because edematous tissue appears hyperintense on diffusion-weighted MR images (13,80). This effect is more pronounced with lower b values (<800 sec/mm²) and can be differentiated from true restriction by visualizing maintained hyperintensity on the corresponding apparent diffusion coefficient map.

In radiation-based treatments, prolonged intervals exceeding 6–8 weeks between baseline imaging and treatment can limit response assessment. Undocumented tumor growth during this period can manifest as apparent enlargement on early posttreatment images, leading to incorrect categorization as LR-TR viable (40,81). In these cases, the LR-TR nonprogressing category should be assigned to allow additional time for a clearer evaluation of tumor response.

Frequently Asked Questions

Can LI-RADS TR assessment criteria be applied to non-HCC malignancies treated with locoregional therapy?

Yes, LI-RADS TR assessment criteria are applicable to non-HCC malignancies treated with locoregional therapy in patients who meet LI-RADS criteria (ie, the presence of cirrhosis, chronic hepatitis B infection, or personal history

Table 2: Pitfalls of the LI-RADS TR Assessment Algorithm and How to Avoid Them		
Pitfall	Why Pitfall Occurs	How To Avoid
Arterial phase mistiming	Most HCCs display peak enhancement during the narrow late arterial phase window, with some lesions enhancing exclusively during this phase	Evaluate portal vein enhancement: early arterial phase images show no portal enhancement, while late arterial phase images show some portal enhancement Recommend short-term follow-up (1–3 months) imaging; consider bolus tracking for CT or MRI and multiarterial phase acquisition for MRI
Perfusional changes simulating viable tumor	Posttreatment inflammatory reactions from nonradiation locoregional therapy, elevated sinusoidal pressure from radiation locoregional therapy, and portal hypoperfusion from portal vein thrombosis can generate arterial hyperenhancement that can appear similar to viable tumor	Look for wedge-shaped and geographic areas of arterial phase hyperenhancement, occasionally round or oval, lacking masslike features If uncertain, assign LR-TR equivocal or LR-TR nonprogressing for follow-up, as perfusional changes resolve over time
Coagulative necrosis masking or simulating viable tumor	Cell death produces coagulative necrosis, showing increased attenuation on CT images and masslike intrinsic hyperintense components on T1-weighted MR images that can simulate or obscure true enhancement	Compare pre- and postcontrast images to identify signal intensity changes; subtraction MRI can be used to distinguish true enhancement from intrinsic hyperintensity
Retained ethiodized oil (Lipiodol; Guerbet) limiting enhancement assessment at CT	Lipiodol increases parenchymal attenuation on CT images, reducing contrast with enhancing tissue; dense oil retention precludes evaluation of enhancement in the treated area	Compare unenhanced and postcontrast images to differentiate retained oil; assign LR-TR nonevaluable when assessment is precluded; consider MRI as an alternative modality
Misinterpretation of diffusion restriction and mild-to-moderate hyperintensity on T2-weighted MR images	Lesions with complete necrosis can still maintain restricted diffusion and mild-to-moderate hyperintensity on T2-weighted MR images; benign causes such as intralesional hemorrhage and liquefactive necrosis in the treated area can produce diffusion restriction	Ensure ancillary features colocalize to areas of equivocal enhancement areas after nonradiation locoregional therapy and to stable or decreasing areas of masslike enhancement after radiation-based locoregional therapy
Inaccurate lesion measurement due to obscured margins	Diffuse HCC margins can be obscured by perfusional alterations or poorly defined when tumor enhancement is equal to that of the background liver	When margins are obscured, evaluate postcontrast images alongside unenhanced T1- or T2-weighted MR images to better delineate tumor margins
Prolonged baseline-to-treatment interval in radiation-based therapy	Undocumented tumor growth during prolonged intervals can be seen as apparent enlargement of the treated lesion	Categorize the lesion as LR-TR nonprogressing for additional follow-up imaging and clearer response evaluation Consider using intraprocedural SPECT/CT from transarterial radioembolization or SBRT simulation images, if available, to provide a more accurate baseline measurement
Misinterpretation of minor size differences for categorizing radiation TR	Minor differences in size may reflect measurement variability rather than actual tumor progression	Assess changes over serial imaging examinations rather than isolated measurements; if unsure, assign LR-TR nonprogressing for additional follow-up imaging

of HCC), because locoregional therapy induces tumor cell death through equivalent mechanisms across tumor types. Although non-HCC malignancies typically exhibit enhancement patterns that differ from those of HCC, the use of LI-RADS criteria still allows assessment of TR because the degree of enhancement and the phase in which enhancement occurs are not relevant, as long as the enhancement is masslike.

How do I assess viability in a lesion that was treated with both radiation- and nonradiation-based locoregional therapy sequentially?

In most cases, the TR assessment algorithm used should be based on the most recently used locoregional therapy. For example, if after nonradiation locoregional therapy the lesion is

categorized as LR-TR viable and is subsequently treated with radiation-based locoregional therapy, then the radiation TR assessment algorithm should be used.

How can a residual or recurrent tumor be differentiated from a new tumor adjacent to a treated lesion?

A residual or recurrent tumor manifests as an area of masslike enhancement in the lesion or at its margin. Conversely, a new tumor is seen separate from the treated lesion.

Can LI-RADS TR assessment criteria be applied to contrast-enhanced US?

New LI-RADS TR assessment criteria for nonradiation therapy were released in 2024. The algorithm has different criteria for

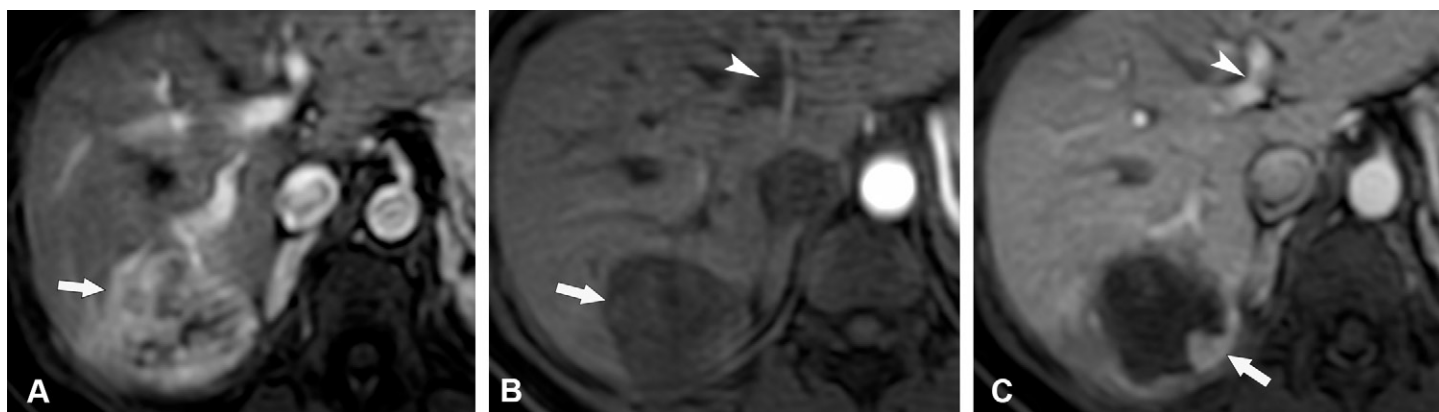


Figure 19. Effect of suboptimal arterial phase timing after TACE in a 51-year-old woman with cirrhosis. **(A)** Pretreatment axial arterial phase MR image shows a 52-mm LI-RADS 5 observation in segment VI (arrow). **(B)** Axial early arterial phase MR image acquired 3 months after TACE shows a heterogeneous treated lesion without apparent masslike enhancement (arrow). Contrast enhancement is limited to the hepatic arteries, with no portal venous enhancement (arrowhead). **(C)** Axial late arterial phase MR image shows irregular masslike enhancement at the treated margin (arrow) that was not apparent on the early phase image, which was categorized as LR-TR viable. Proper timing of the late arterial phase is evident from portal vein enhancement exceeding that of the liver parenchyma (arrowhead).

tumor viability, reflecting specific properties of contrast-enhanced US as a modality (11).

Does LR-TR nonviable exclude microscopic viability?

No, LR-TR nonviable indicates a low likelihood of clinically significant viable tumor tissue, but imaging cannot exclude microscopic foci of viable cells. Current evidence demonstrates that the LR-TR nonviable criteria accurately predict complete tumor necrosis in pathologic studies, with negative predictive values of 81%–93% (28,30,39). However, radiologic detection requires a minimal tumor volume for reliable assessment. A study published in 2022 (29) demonstrated 35% sensitivity for the LR-TR viable or equivocal category in detecting tumors smaller than 10 mm, compared with 80% for tumors measuring at least 10 mm.

Conclusion

Multiple locoregional therapies are available for HCC, with imaging findings specific to each modality. The updated 2024 CT and MRI LI-RADS TR assessment criteria provided distinct algorithms for use after nonradiation or radiation-based therapies, simplified the definition of viable disease, and incorporated MRI ancillary features to enhance detection of tumor viability. These updated criteria can improve diagnostic accuracy, resulting in more effective clinical management of HCC.

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References

- Toh MR, Wong EYT, Wong SH, et al. Global Epidemiology and Genetics of Hepatocellular Carcinoma. *Gastroenterology* 2023;164(5):766–782.
- European Association for the Study of the Liver. EASL Clinical Practice Guidelines on the management of hepatocellular carcinoma. *J Hepatol* 2025;82(2):315–374.
- Singal AG, Llovet JM, Yatchew M, et al. AASLD Practice Guidance on prevention, diagnosis, and treatment of hepatocellular carcinoma. *Hepatology* 2023;78(6):1922–1965. [Published correction appears in *Hepatology* 2023;78(6):E105.]
- Chernyak V, Fowler KJ, Kamaya A, et al. Liver Imaging Reporting and Data System (LI-RADS) Version 2018: Imaging of Hepatocellular Carcinoma in At-Risk Patients. *Radiology* 2018;289(3):816–830.
- Riaz A, Lewandowski RJ, Kulik L, et al. Radiologic-pathologic correlation of hepatocellular carcinoma treated with chemoembolization. *Cardiovasc Intervent Radiol* 2010;33(6):1143–1152.
- Morimoto M, Sugimori K, Shirato K, et al. Treatment of hepatocellular carcinoma with radiofrequency ablation: radiologic-histologic correlation during follow-up periods. *Hepatology* 2002;35(6):1467–1475.
- Mai Z, Yang Q, Xu J, et al. Response evaluation of hepatocellular carcinoma treated with stereotactic body radiation therapy: magnetic resonance imaging findings. *Abdom Radiol (NY)* 2023;48(6):1995–2007.
- Mendiratta-Lala M, Masch W, Shankar PR, et al. Magnetic Resonance Imaging Evaluation of Hepatocellular Carcinoma Treated With Stereotactic Body Radiation Therapy: Long Term Imaging Follow-Up. *Int J Radiat Oncol Biol Phys* 2019;103(1):169–179.
- Kampalath R, Mendiratta-Lala M, Lewis S, Benefield T, Yaghamai V, Burke L. Society of abdominal radiology survey of practice patterns in using LI-RADS treatment response criteria in the evaluation of hepatocellular carcinoma post-locoregional treatment. *Abdom Radiol (NY)* 2023;48(11):3401–3407.
- American College of Radiology 2018 Liver Imaging Reporting and Data Systems, version 2018. <https://www.acr.org/Clinical-Resources/Clinical-Tools-and-Reference/Reporting-and-Data-Systems/LI-RADS>. Accessed February 23, 2025.
- Lyshchik A, Fetzter DT, Kono Y, et al. Liver Imaging Reporting and Data System Contrast-Enhanced US Nonradiation Treatment Response Assessment Version 2024. *Radiology* 2024;311(2):e232369.
- Yacoub JH, Elsayes KM, Fowler KJ, et al. Pitfalls in liver MRI: Technical approach to avoiding misdiagnosis and improving image quality. *J Magn Reson Imaging* 2019;49(1):41–58.

13. Kamal O, Roudenko A, Diab M, et al. Common pitfalls and diagnostic challenges in the application of LI-RADS CT/MRI algorithms: a comprehensive review. *Abdom Radiol (NY)* 2025;50(7):2944–2957.
14. Ahmed M, Solbiati L, Brace CL, et al; Standard of Practice Committee of the Cardiovascular and Interventional Radiological Society of Europe. Image-guided tumor ablation: standardization of terminology and reporting criteria—a 10-year update. *Radiology* 2014;273(1):241–260.
15. Gupta P, Maralakunte M, Kumar M P, et al. Overall survival and local recurrence following RFA, MWA, and cryoablation of very early and early HCC: a systematic review and Bayesian network meta-analysis. *Eur Radiol* 2021;31(7):5400–5408.
16. Kwan SW, Fidelman N, Ma E, Kerlan RK Jr, Yao FY. Imaging predictors of the response to transarterial chemoembolization in patients with hepatocellular carcinoma: a radiological-pathological correlation. *Liver Transpl* 2012;18(6):727–736.
17. Llovet JM, Bruix J. Systematic review of randomized trials for unresectable hepatocellular carcinoma: Chemoembolization improves survival. *Hepatology* 2003;37(2):429–442.
18. Cammà C, Schepis F, Orlando A, et al. Transarterial chemoembolization for unresectable hepatocellular carcinoma: meta-analysis of randomized controlled trials. *Radiology* 2002;224(1):47–54.
19. Golfieri R, Giampalma E, Renzulli M, et al. Randomised controlled trial of doxorubicin-eluting beads vs conventional chemoembolisation for hepatocellular carcinoma. *Br J Cancer* 2014;111(2):255–264.
20. Brown KT, Do RK, Gonen M, et al. Randomized Trial of Hepatic Artery Embolization for Hepatocellular Carcinoma Using Doxorubicin-Eluting Microspheres Compared With Embolization With Microspheres Alone. *J Clin Oncol* 2016;34(17):2046–2053.
21. Kim S, Mannelli L, Hajdu CH, et al. Hepatocellular carcinoma: assessment of response to transarterial chemoembolization with image subtraction. *J Magn Reson Imaging* 2010;31(2):348–355.
22. Dromain C, de Baere T, Elias D, et al. Hepatic tumors treated with percutaneous radio-frequency ablation: CT and MR imaging follow-up. *Radiology* 2002;223(1):255–262.
23. Lim HK, Choi D, Lee WJ, et al. Hepatocellular carcinoma treated with percutaneous radio-frequency ablation: evaluation with follow-up multiphase helical CT. *Radiology* 2001;221(2):447–454.
24. Kim KW, Lee JM, Choi BI. Assessment of the treatment response of HCC. *Abdom Imaging* 2011;36(3):300–314.
25. Mendiratta-Lala M, Aslam A, Bai HX, et al. Ethiodized oil as an imaging biomarker after conventional transarterial chemoembolization. *Eur Radiol* 2024;34(5):3284–3297.
26. Kierans AS, Najjar M, Dutruel SP, et al. Evaluation of the LI-RADS treatment response algorithm in hepatocellular carcinoma after trans-arterial chemoembolization. *Clin Imaging* 2021;80:117–122.
27. Yoon J, Lee S, Shin J, Kim SS, Kim GM, Won JY. LI-RADS Version 2018 Treatment Response Algorithm: Diagnostic Performance after Transarterial Radioembolization for Hepatocellular Carcinoma. *Korean J Radiol* 2021;22(8):1279–1288.
28. Shropshire EL, Chaudhry M, Miller CM, et al. LI-RADS Treatment Response Algorithm: Performance and Diagnostic Accuracy. *Radiology* 2019;292(1):226–234.
29. Kim YY, Kim MJ, Yoon JK, Shin J, Roh YH. Incorporation of Ancillary MRI Features Into the LI-RADS Treatment Response Algorithm: Impact on Diagnostic Performance After Locoregional Treatment of Hepatocellular Carcinoma. *AJR Am J Roentgenol* 2022;218(3):484–493.
30. Chaudhry M, McGinty KA, Mervak B, et al. The LI-RADS Version 2018 MRI Treatment Response Algorithm: Evaluation of Ablated Hepatocellular Carcinoma. *Radiology* 2020;294(2):320–326.
31. Zhang Y, Wang J, Li H, et al. Performance of LI-RADS version 2018 CT treatment response algorithm in tumor response evaluation and survival prediction of patients with single hepatocellular carcinoma after radiofrequency ablation. *Ann Transl Med* 2020;8(6):388.
32. Ormiston WEL, Yarmohammadi H, Lobaugh S, et al. Post-treatment CT LI-RADS categories: predictors of overall survival in hepatocellular carcinoma post bland transarterial embolization. *Abdom Radiol (NY)* 2021;46(8):3738–3747.
33. Hsin IF, Hsu CY, Huang HC, et al. Liver failure after transarterial chemoembolization for patients with hepatocellular carcinoma and ascites: incidence, risk factors, and prognostic prediction. *J Clin Gastroenterol* 2011;45(6):556–562.
34. Cheng KL, Cheng YM, Chan CY, Wang CC. Predictors of Liver Dysfunction After Transhepatic Arterial Chemo-embolization in Hepatocellular Carcinoma Patients. *Dig Dis Sci* 2023;68(8):3467–3472.
35. Kim TH, Woo S, Joo I, et al. LI-RADS treatment response algorithm for detecting incomplete necrosis in hepatocellular carcinoma after locoregional treatment: a systematic review and meta-analysis using individual patient data. *Abdom Radiol (NY)* 2021;46(8):3717–3728.
36. Huh YJ, Kim DH, Kim B, Choi JI, Rha SE. Per-Feature Accuracy of Liver Imaging Reporting and Data System Locoregional Treatment Response Algorithm: A Systematic Review and Meta-Analysis. *Cancers (Basel)* 2021;13(17):4432.
37. Cools KS, Moon AM, Burke LMB, McGinty KA, Strassle PD, Gerber DA. Validation of the Liver Imaging Reporting and Data System Treatment Response Criteria After Thermal Ablation for Hepatocellular Carcinoma. *Liver Transpl* 2020;26(2):203–214.
38. Park S, Joo I, Lee DH, et al. Diagnostic Performance of LI-RADS Treatment Response Algorithm for Hepatocellular Carcinoma: Adding Ancillary Features to MRI Compared with Enhancement Patterns at CT and MRI. *Radiology* 2020;296(3):554–561.
39. Kim SW, Joo I, Kim HC, et al. LI-RADS treatment response categorization on gadoteric acid-enhanced MRI: diagnostic performance compared to mRECIST and added value of ancillary features. *Eur Radiol* 2020;30(5):2861–2870.
40. Aslam A, Chernyak V, Tang A, et al. CT/MRI LI-RADS 2024 Update: Treatment Response Assessment. *Radiology* 2024;313(2):e232408.
41. Watanabe H, Kanematsu M, Goshima S, et al. Is gadoterate disodium-enhanced MRI useful for detecting local recurrence of hepatocellular carcinoma after radiofrequency ablation therapy? *AJR Am J Roentgenol* 2012;198(3):589–595.
42. Zhou S, Zhou G, Shen Y, et al. LI-RADS Nonradiation Treatment Response Algorithm Version 2024: Diagnostic Performance and Impact of Ancillary Features. *AJR Am J Roentgenol* 2025;224(2):e2432035.
43. Mulé S, Colosio A, Cazejust J, Kianmanesh R, Soyer P, Hoeffel C. Imaging of the postoperative liver: review of normal appearances and common complications. *Abdom Imaging* 2015;40(7):2761–2776.
44. Arrivé L, Hricak H, Goldberg HI, Thoeni RF, Margulis AR. MR appearance of the liver after partial hepatectomy. *AJR Am J Roentgenol* 1989;152(6):1215–1220.
45. Chen H, Han Z, Luo Q, et al. Radiotherapy modulates tumor cell fate decisions: a review. *Radiat Oncol* 2022;17(1):196.
46. Eriksson D, Stigbrand T. Radiation-induced cell death mechanisms. *Tumour Biol* 2010;31(4):363–372.
47. Navin PJ, Olson MC, Mendiratta-Lala M, Hallemeier CL, Torbenson MS, Venkatesh SK. Imaging Features in the Liver after Stereotactic Body Radiation Therapy. *RadioGraphics* 2022;42(7):2131–2148.
48. Gudkov AV, Komarova EA. The role of p53 in determining sensitivity to radiotherapy. *Nat Rev Cancer* 2003;3(2):117–129.
49. Lewis S, Dawson L, Barry A, Stanescu T, Mohamad I, Hosni A. Stereotactic body radiation therapy for hepatocellular carcinoma: From infancy to ongoing maturity. *JHEP Rep* 2022;4(8):100498.
50. Miller FH, Lopes Vendrami C, Gabr A, et al. Evolution of Radioembolization in Treatment of Hepatocellular Carcinoma: A Pictorial Review. *RadioGraphics* 2021;41(6):1802–1818.
51. Salem R, Padia SA, Lam M, et al. Clinical, dosimetric, and reporting considerations for Y-90 glass microspheres in hepatocellular carcinoma: updated 2022 recommendations from an international multidisciplinary working group. *Eur J Nucl Med Mol Imaging* 2023;50(2):328–343.
52. Lam M, Garin E, Maccauro M, et al. A global evaluation of advanced dosimetry in transarterial radioembolization of hepatocellular carcinoma with Yttrium-90: the TARGET study. *Eur J Nucl Med Mol Imaging* 2022;49(10):3340–3352.
53. Watanabe M, Grafe H, Theysohn J, et al. Voxel-Based Dosimetry Predicts Hepatotoxicity in Hepatocellular Carcinoma Patients Undergoing Radioembolization with 90Y Glass Microspheres. *J Nucl Med* 2023;64(7):1102–1108.
54. Salem R, Johnson GE, Kim E, et al. Yttrium-90 Radioembolization for the Treatment of Solitary, Unresectable HCC: The LEGACY Study. *Hepatology* 2021;74(5):2342–2352.
55. Sarwar A, Malik MS, Vo NH, et al. Efficacy and Safety of Radiation Segmentectomy with 90Y Resin Microspheres for Hepatocellular Carcinoma. *Radiology* 2024;311(2):e231386.
56. Rim CH, Kim HJ, Seong J. Clinical feasibility and efficacy of stereotactic body radiotherapy for hepatocellular carcinoma: A systematic review and meta-analysis of observational studies. *Radiother Oncol* 2019;131:135–144.
57. Mastrocostas K, Fischer S, Munoz-Schuffenegger P, et al. Radiological tumor response and histopathological correlation of hepatocellular carcinoma treated with stereotactic body radiation therapy as a bridge to liver transplantation. *Abdom Radiol (NY)* 2021;46(4):1572–1585.
58. Mendiratta-Lala M, Aslam A, Maturen KE, et al. LI-RADS Treatment Response Algorithm: Performance and Diagnostic Accuracy With Radiologic-Pathologic Explant Correlation in Patients With SBRT-Treated Hepatocellular Carcinoma. *Int J Radiat Oncol Biol Phys* 2022;112(3):704–714.
59. Stocker D, King MJ, Homs ME, et al. Early post-treatment MRI predicts long-term hepatocellular carcinoma response to radiation segmentectomy. *Eur Radiol* 2024;34(1):475–484.

60. Sanuki N, Takeda A, Mizuno T, et al. Tumor response on CT following hypofractionated stereotactic ablative body radiotherapy for small hypervascular hepatocellular carcinoma with cirrhosis. *AJR Am J Roentgenol* 2013;201(6):W812–W820.
61. Wei H, Jiang H, Yoo J, et al. Temporal evolution of the LI-RADS radiation treatment response assessment on multiphase CT/MRI in patients undergoing selective internal radiation therapy for hepatocellular carcinoma. *Eur Radiol* 2025. 10.1007/s00330-025-11659-1. Published online May 17, 2025.
62. Chiu SH, Kesselman A, Yoon L, Kamaya A, Tse JR. LI-RADS CT/MRI Radiation Treatment Response Algorithm Version 2024: Category Redistribution and Short-Term Outcomes in Patients Undergoing Y-90 Radioembolization for HCC. *AJR Am J Roentgenol* 2025. 10.2214/AJR.25.32745. Published online April 30, 2025.
63. Seyal AR, Gonzalez-Guindalini FD, Arslanoglu A, et al. Reproducibility of mRECIST in assessing response to transarterial radioembolization therapy in hepatocellular carcinoma. *Hepatology* 2015;62(4):1111–1121.
64. Yoon SH, Kim KW, Goo JM, Kim DW, Hahn S. Observer variability in RECIST-based tumour burden measurements: a meta-analysis. *Eur J Cancer* 2016;53:5–15.
65. Vietti Violi N, Gnerre J, Law A, et al. Assessment of HCC response to Yttrium-90 radioembolization with gadoxetate disodium MRI: correlation with histopathology. *Eur Radiol* 2022;32(9):6493–6503.
66. Rhee TK, Naik NK, Deng J, et al. Tumor response after yttrium-90 radioembolization for hepatocellular carcinoma: comparison of diffusion-weighted functional MR imaging with anatomic MR imaging. *J Vasc Interv Radiol* 2008;19(8):1180–1186.
67. Park MJ, Kim SY, Yoon SM, et al. Stereotactic body radiotherapy-induced arterial hypervascularity of non-tumorous hepatic parenchyma in patients with hepatocellular carcinoma: potential pitfalls in tumor response evaluation on multiphase computed tomography. *PLoS One* 2014;9(2):e90327.
68. Miller FH, Keppke AL, Reddy D, et al. Response of liver metastases after treatment with yttrium-90 microspheres: role of size, necrosis, and PET. *AJR Am J Roentgenol* 2007;188(3):776–783.
69. Venkatesh SK, Harper KC, Borhani AA, et al. Hepatic Sinusoidal Disorders. *RadioGraphics* 2024;44(9):e240006.
70. Kim J, Jung Y. Radiation-induced liver disease: current understanding and future perspectives. *Exp Mol Med* 2017;49(7):e359.
71. Takamatsu S, Kozaka K, Kobayashi S, et al. Pathology and images of radiation-induced hepatitis: a review article. *Jpn J Radiol* 2018;36(4):241–256.
72. Lall C, Bhargava P, Sandrasegaran K, et al. Three-dimensional conformal radiation therapy in the liver: MRI findings along a time continuum. *J Comput Assist Tomogr* 2015;39(3):356–364.
73. Rosenkrantz AB, Lee L, Matza BW, Kim S. Infiltrative hepatocellular carcinoma: comparison of MRI sequences for lesion conspicuity. *Clin Radiol* 2012;67(12):e105–e111.
74. Gerena M, Molvar C, Masciocchi M, et al. LI-RADS treatment response assessment of combination locoregional therapy for HCC. *Abdom Radiol (NY)* 2021;46(8):3634–3647.
75. Cannella R, Lewis S, da Fonseca L, Ronot M, Rimola J. Immunotherapy-Based Treatments of Hepatocellular Carcinoma: AJR Expert Panel Narrative Review. *AJR Am J Roentgenol* 2022;219(4):533–546.
76. Masch WR, Kampalath R, Parikh N, Shampain KA, Aslam A, Chernyak V. Imaging of treatment response during systemic therapy for hepatocellular carcinoma. *Abdom Radiol (NY)* 2021;46(8):3625–3633.
77. Bitar R, Salem R, Finn R, Greten TF, Goldberg SN, Chapiro J. Interventional Oncology Meets Immuno-oncology: Combination Therapies for Hepatocellular Carcinoma. *Radiology* 2024;313(2):e232875. [Published correction appears in *Radiology* 2025;314(2):e259004.]
78. Goshima S, Kanematsu M, Matsuo M, et al. Early-enhancing nonneoplastic lesions on gadolinium-enhanced magnetic resonance imaging of the liver following partial hepatectomy. *J Magn Reson Imaging* 2004;20(1):66–74.
79. Kloeckner R, Otto G, Biesterfeld S, Oberholzer K, Dueber C, Pitton MB. MDCT versus MRI assessment of tumor response after transarterial chemoembolization for the treatment of hepatocellular carcinoma. *Cardiovasc Intervent Radiol* 2010;33(3):532–540.
80. Lall C, Bura V, Lee TK, et al. Diffusion-weighted imaging in hemorrhagic abdominal and pelvic lesions: restricted diffusion can mimic malignancy. *Abdom Radiol (NY)* 2018;43(7):1772–1784.
81. Castagnoli F, Mencil J, Ap Dafydd D, et al. Response Evaluation Criteria in Gastrointestinal and Abdominal Cancers: Which to Use and How to Measure. *RadioGraphics* 2024;44(5):e230047.